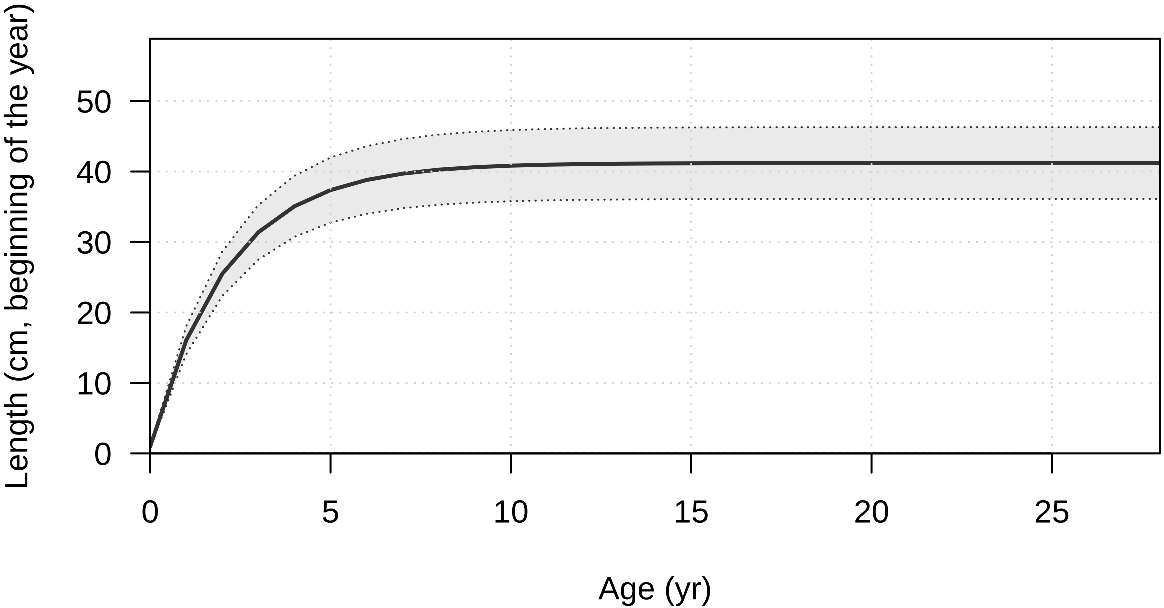
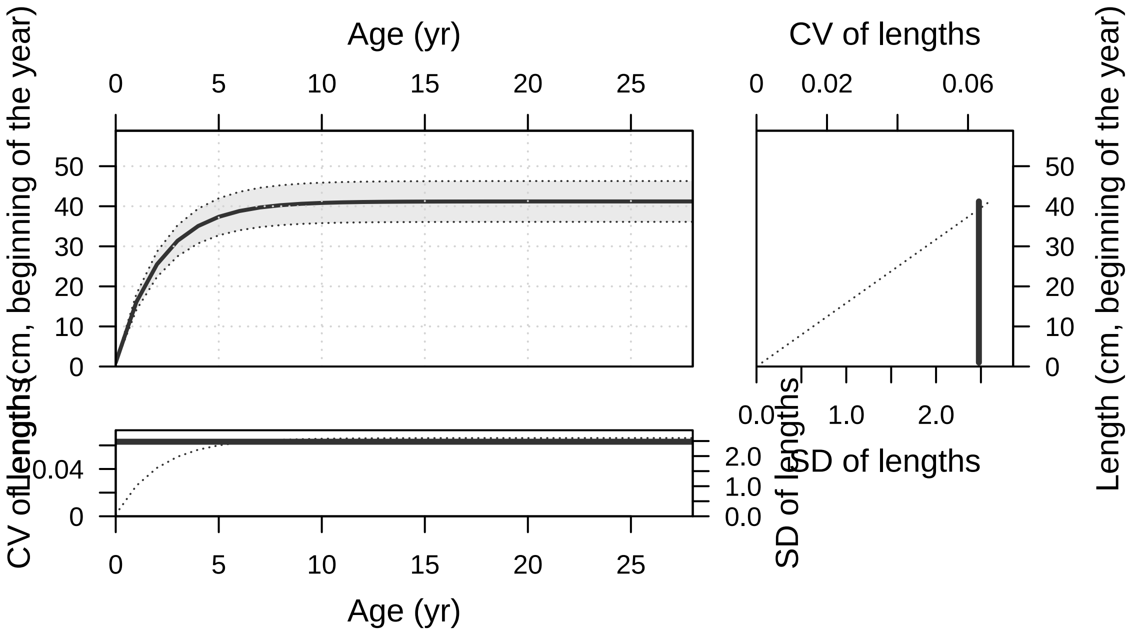
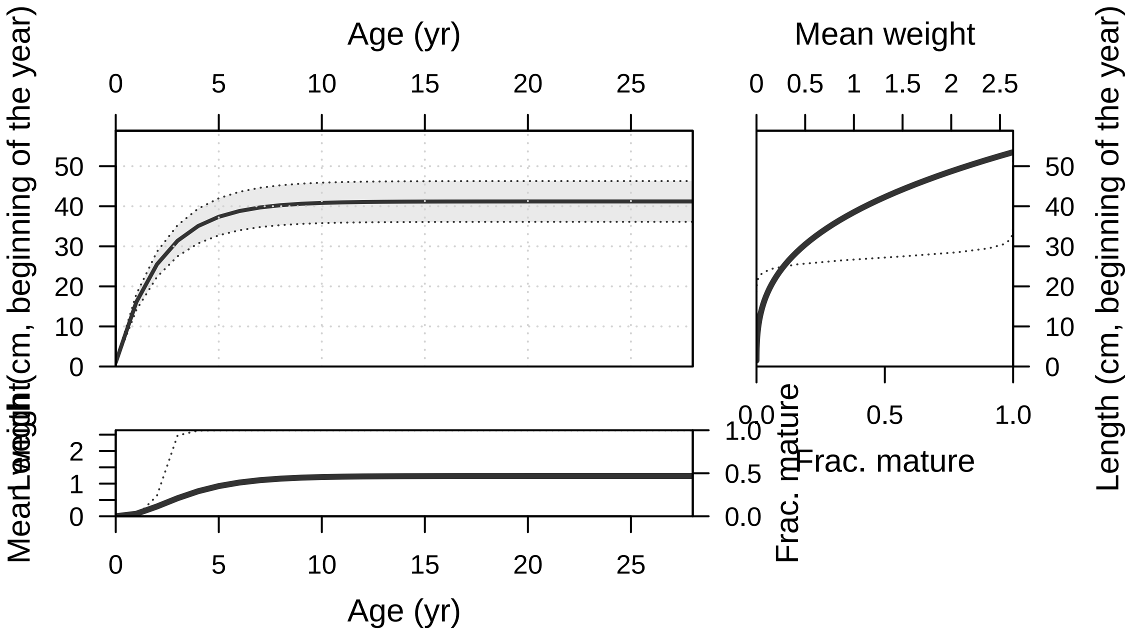
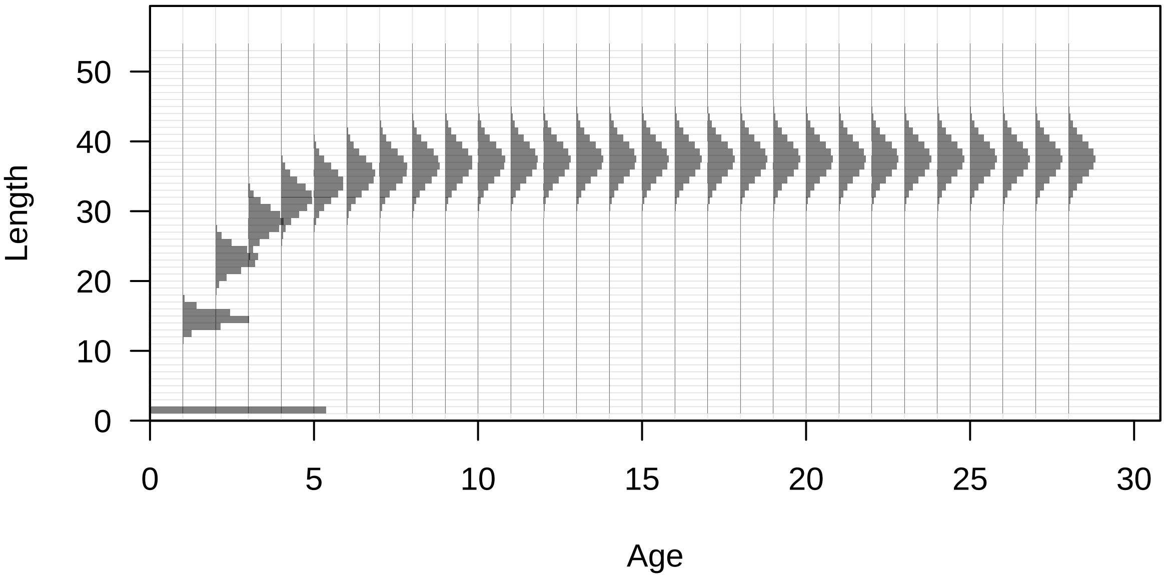
Plots created using the 'r4ss' package in R
Stock Synthesis version: 3.30.19.0
StartTime: Tue Feb 10 10:23:16 2026
Data_File: data.ss
Control_File: control.ss

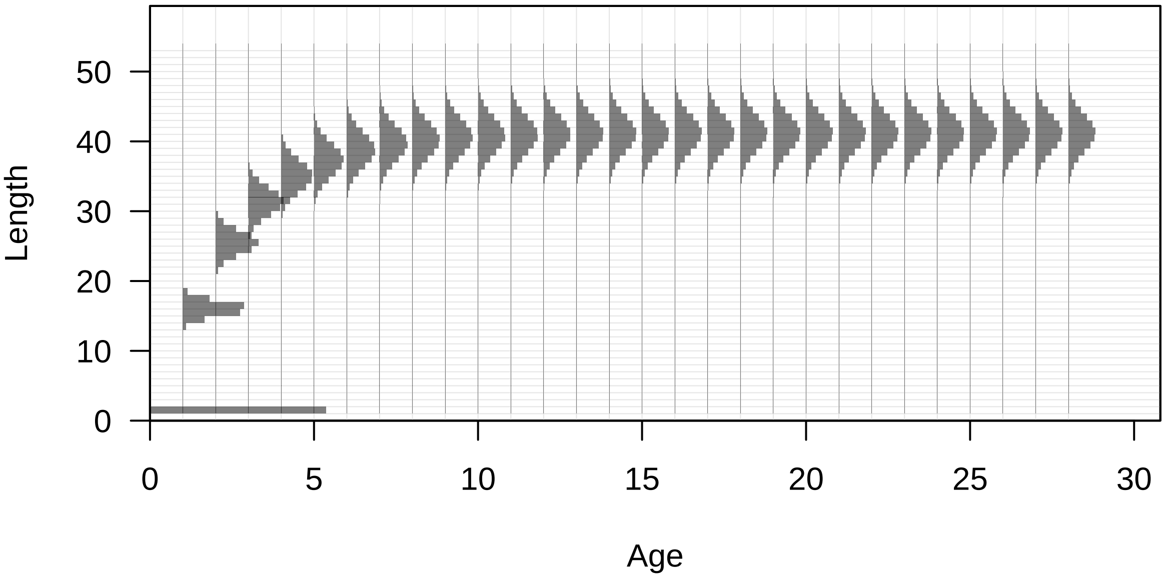

Length (cm, beginning of the year)
0
10
20
30
40
50
0
5
10
15
20
25
Age (yr)

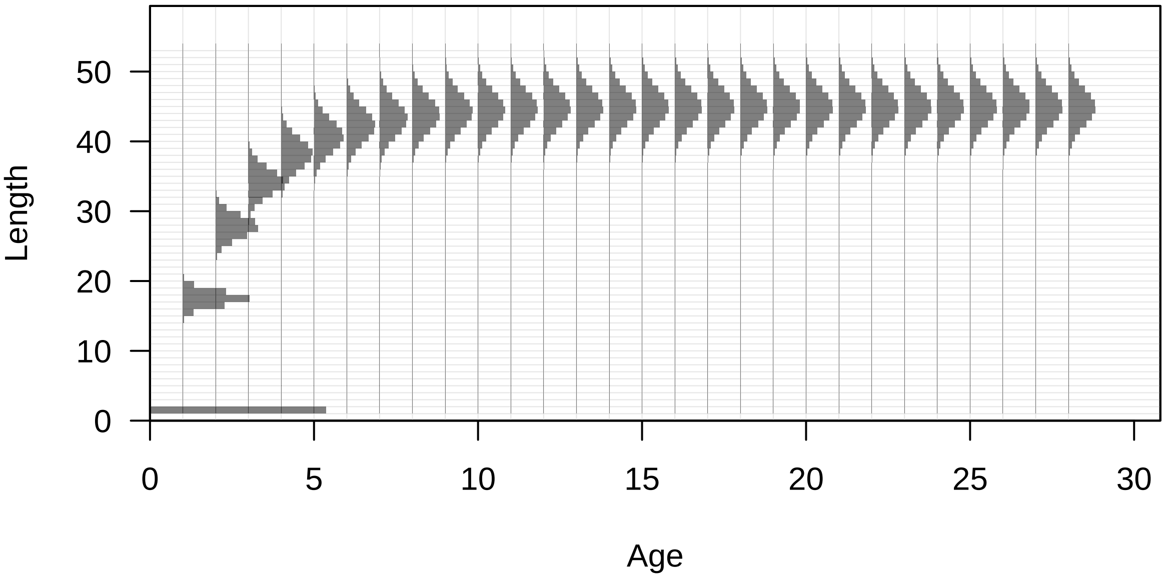


Age (yr)
0
5
10
15
20
25
CV of lengths
0
0.02
0.06
Length (cm, beginning of the year)
50
40
30
20
10
0
0.0
1.0
2.0
SD of lengths
CV of lengths
0.04
0
2.0
1.0
0.0
SD of lengths
Age (yr)

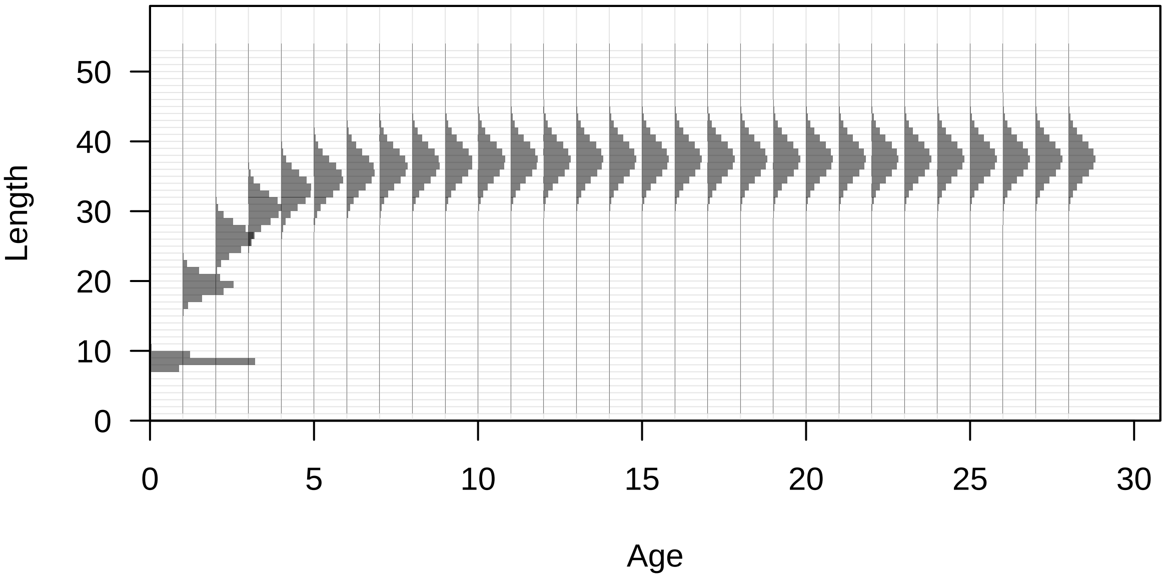

Age (yr)
Mean weight
0
5
10
15
20
25
0
0.5
1
1.5
2
2.5
Length (cm, beginning of the year)
50
40
30
20
10
0
0.0
0.5
1.0
Frac. mature
Mean weight
2
1
0
1.0
0.5
0.0
Frac. mature
Age (yr)
Length (cm, beginning of the year)

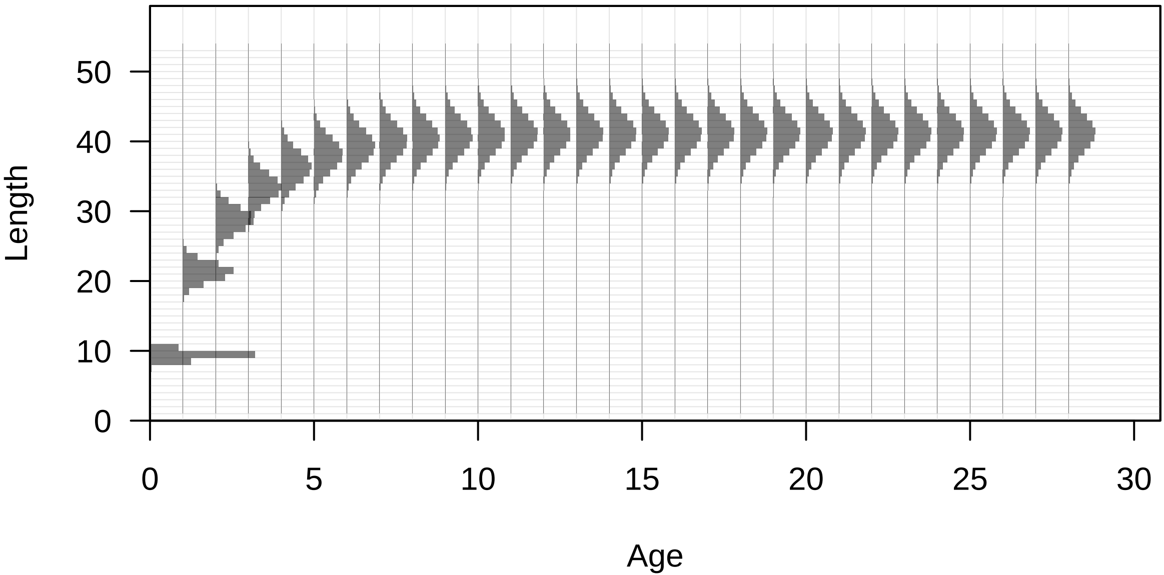

Length
50
40
30
20
10
0
0
5
10
15
20
25
30
Age


Length
Age
0
10
20
30
40
50
0
5
10
15
20
25
30

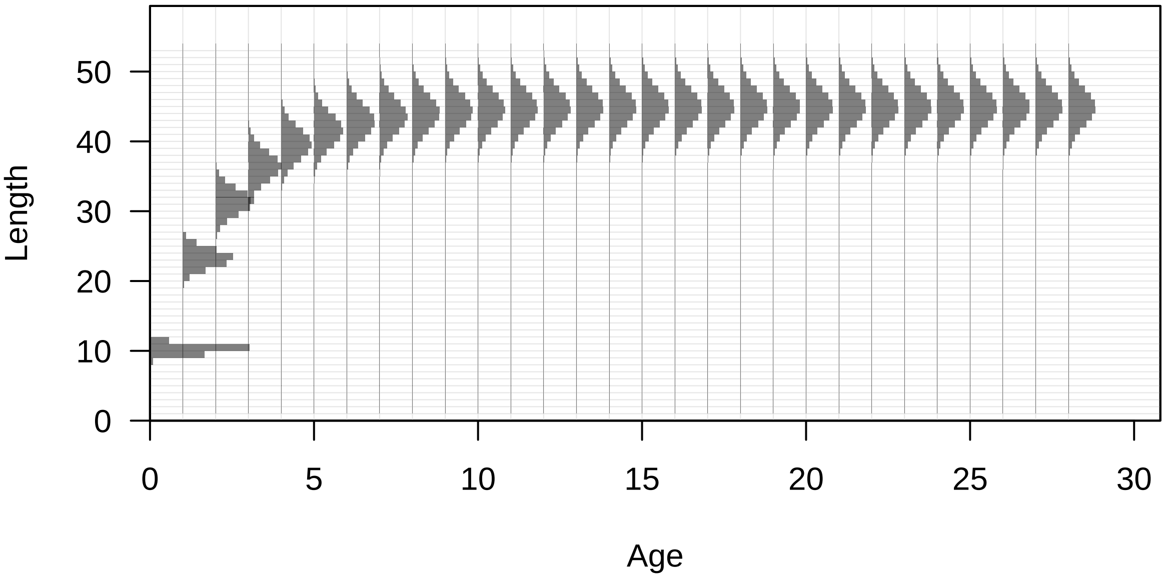


Length
0
10
20
30
40
50
0
5
10
15
20
25
30
Age

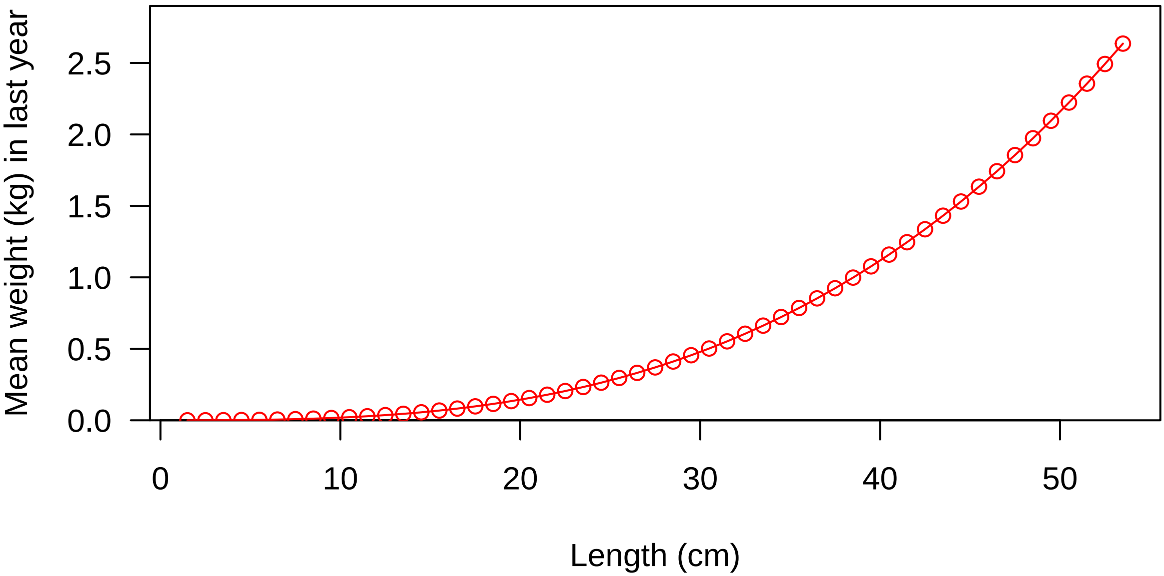

Length
Age
0
10
20
30
40
50
0
5
10
15
20
25
30

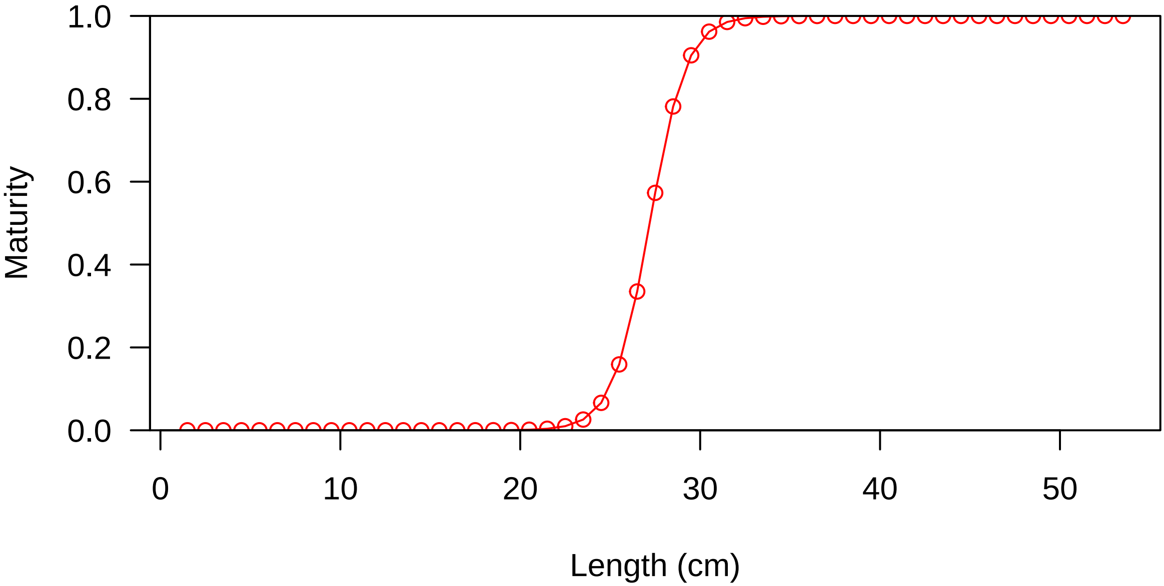


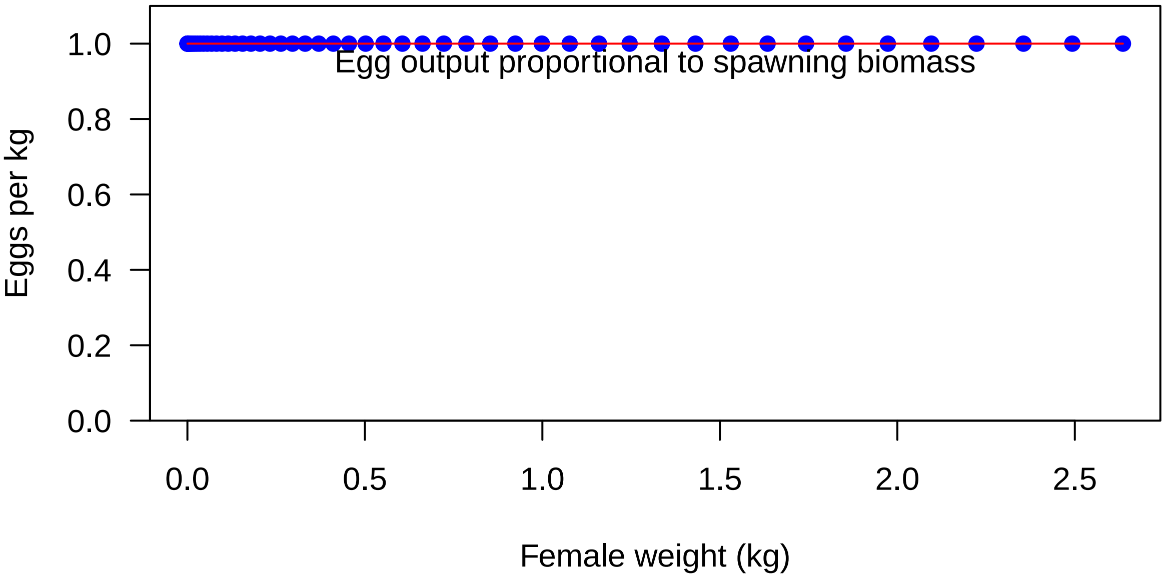
Length
50
40
30
20
10
0
0
5
10
15
20
25
30
Age



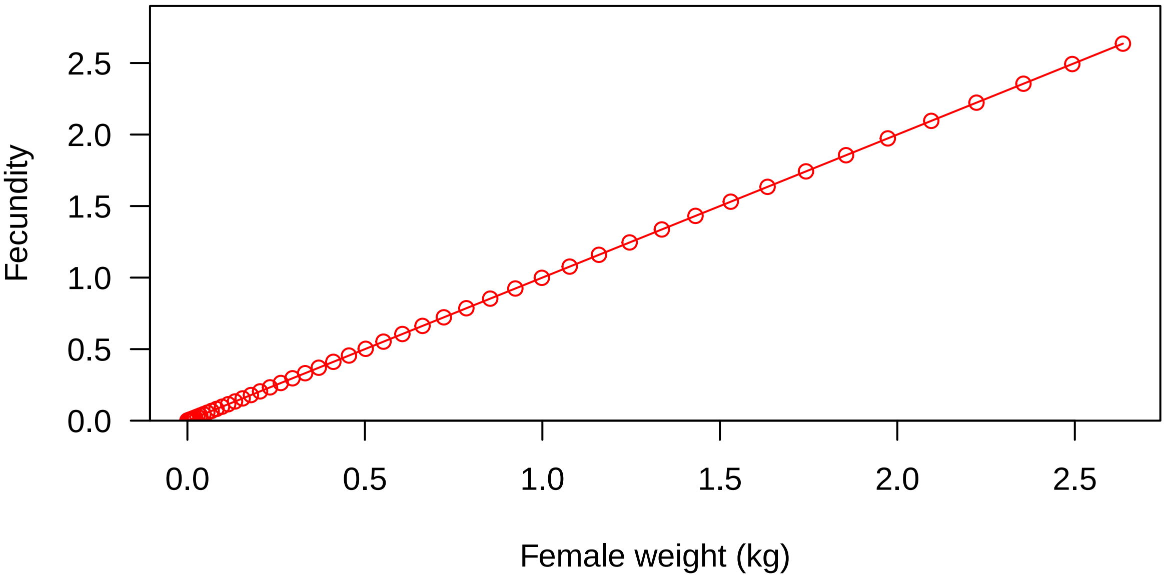
Length
0
10
20
30
40
50
0
5
10
15
20
25
30
Age

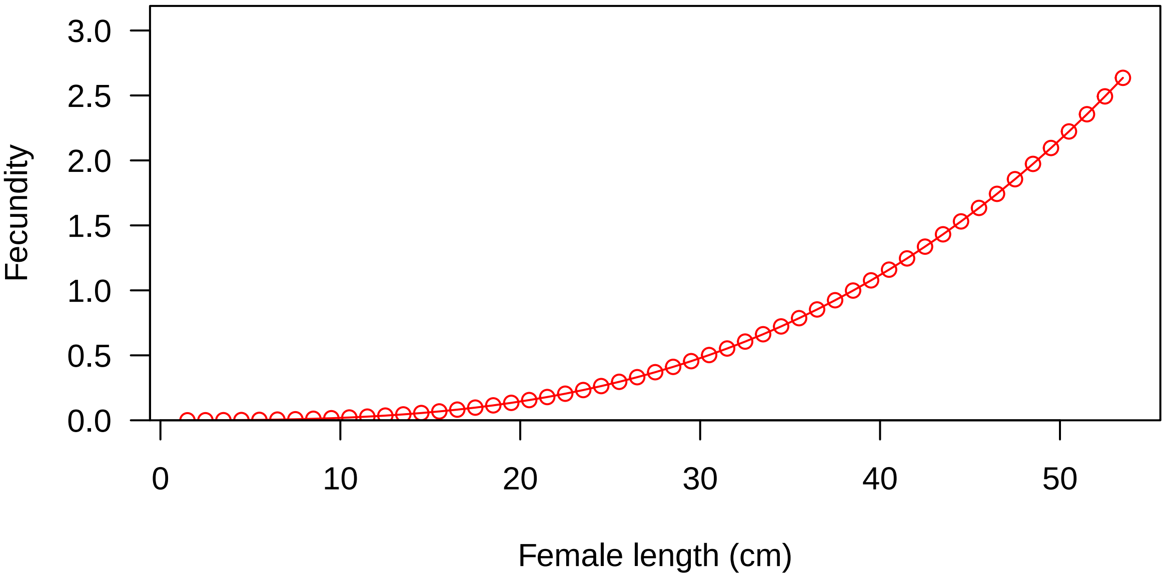

Mean weight (kg) in last year
0.0
0.5
1.0
1.5
2.0
2.5
0
10
20
30
40
50
Length (cm)


1.0
0.8
0.6
0.4
0.2
0.0
Maturity
0
10
20
30
40
50
Length (cm)

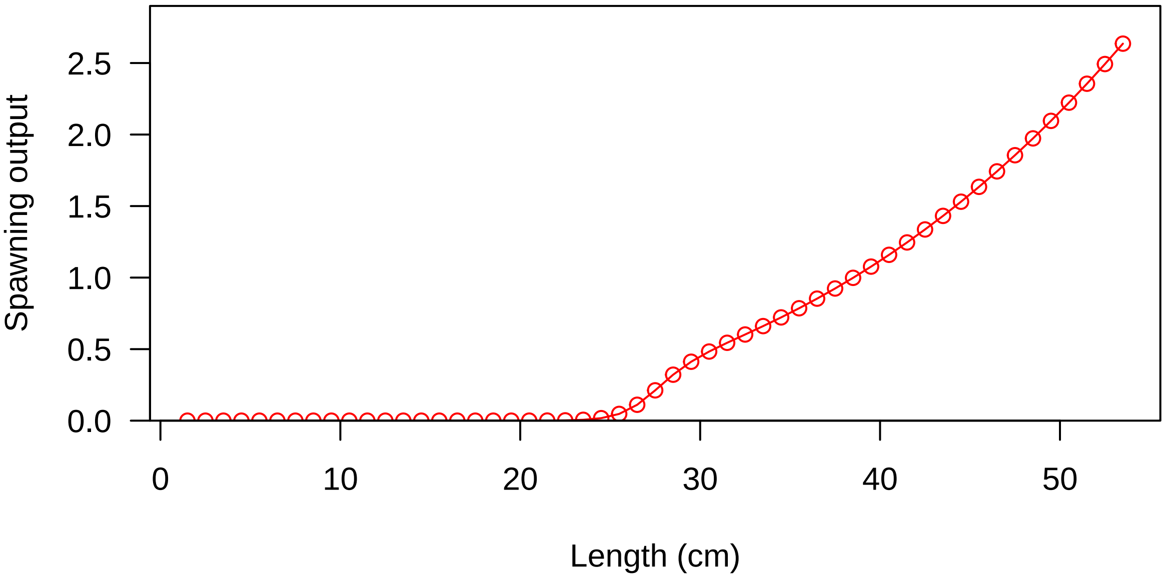


Egg output proportional to spawning biomass
Eggs per kg
Female weight (kg)
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5

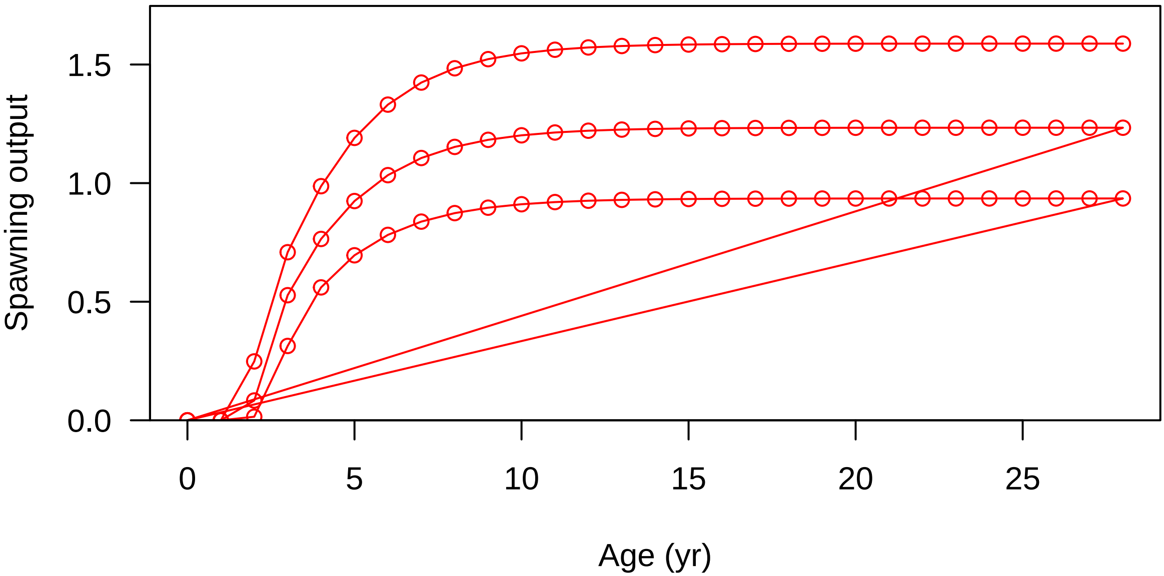

Fecundity
Female weight (kg)
0.0
0.5
1.0
1.5
2.0
2.5



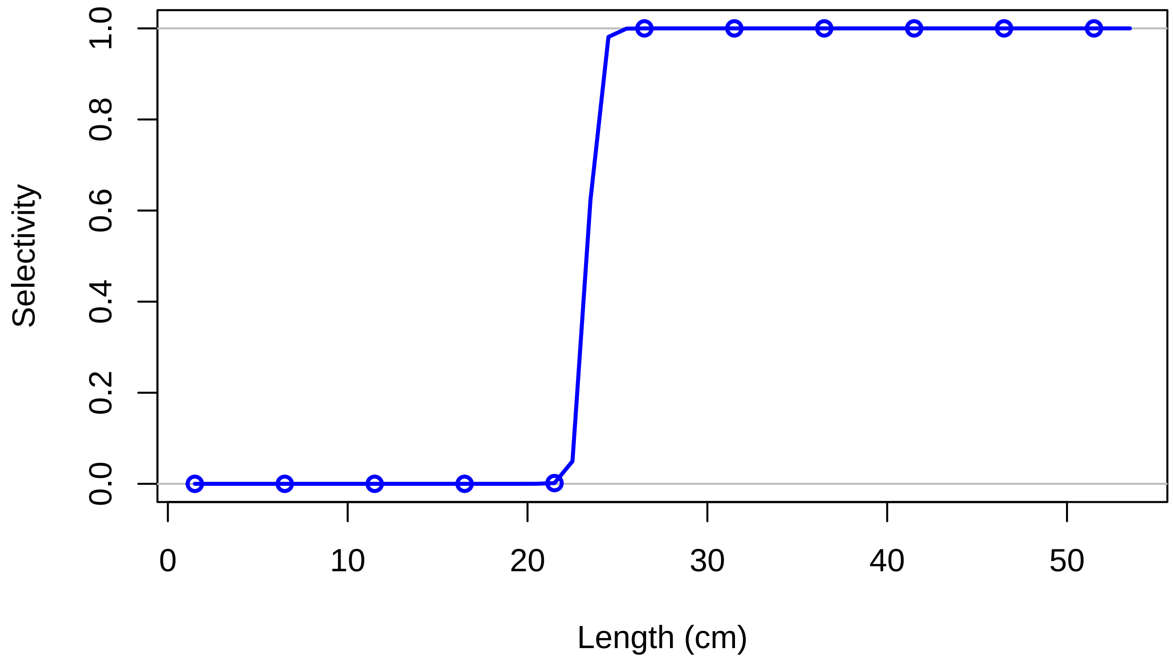
Fecundity
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
50
Female length (cm)



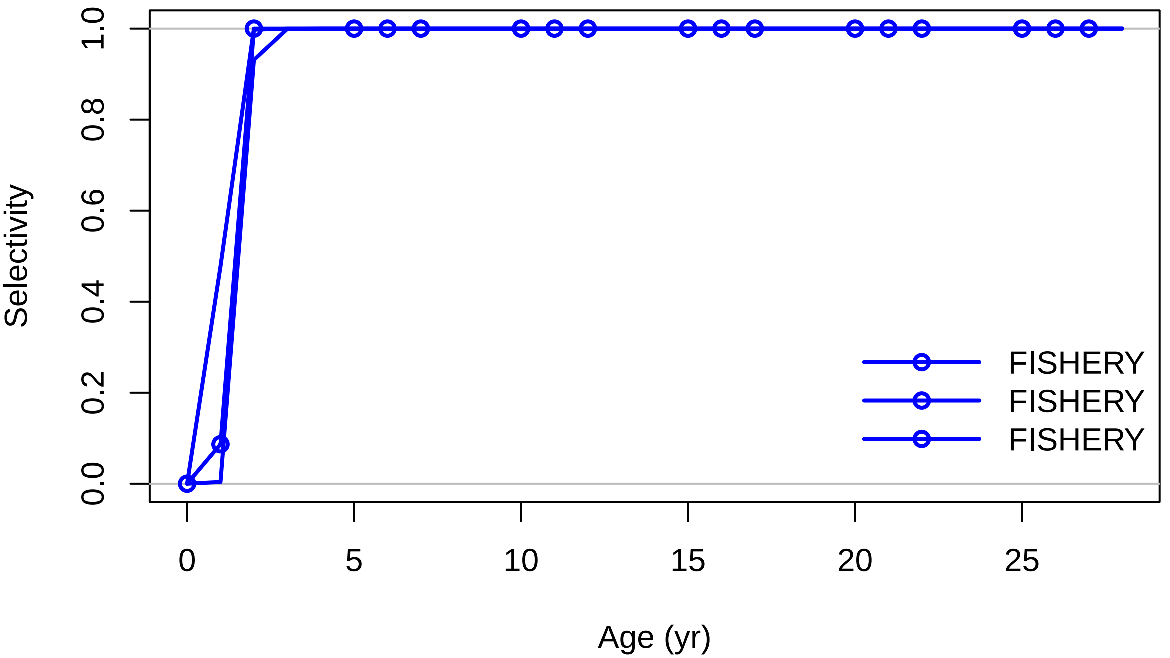
Spawning output
0.0
0.5
1.0
1.5
2.0
2.5
0
10
20
30
40
50
Length (cm)



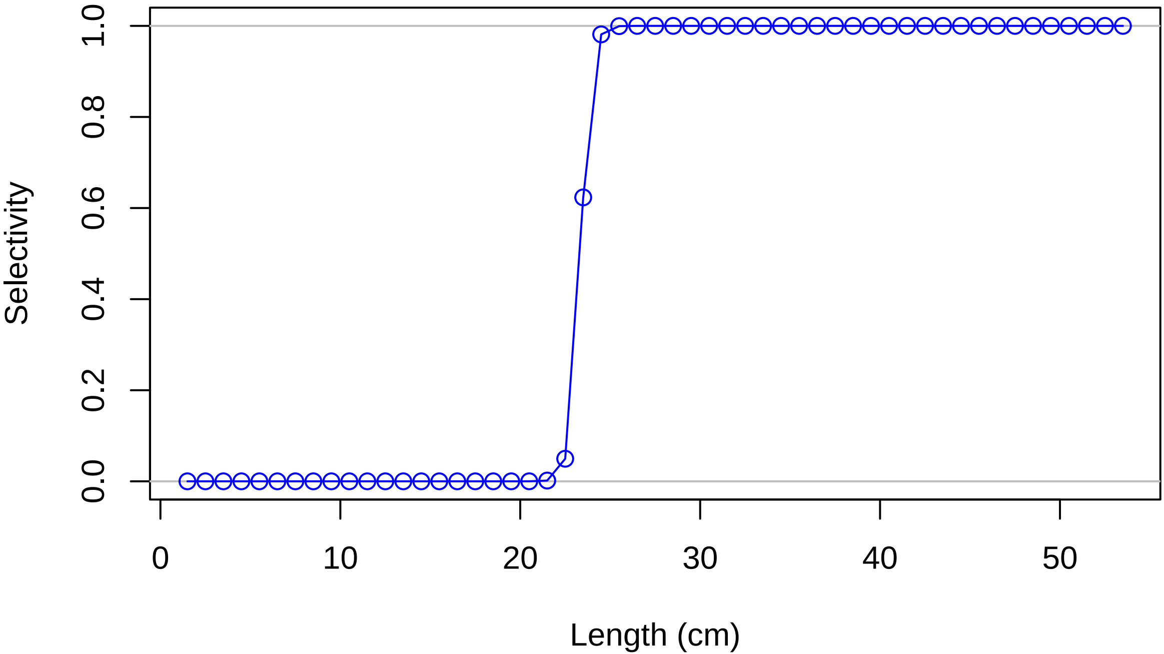
Spawning output
0.0
0.5
1.0
1.5
0
5
10
15
20
25
Age (yr)



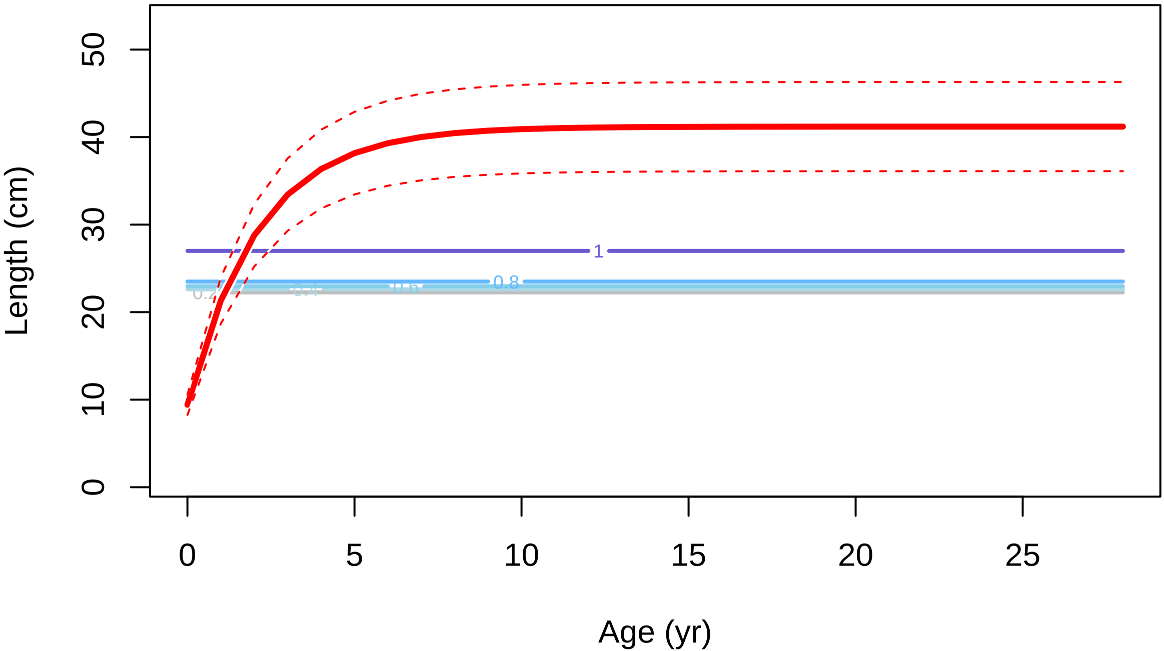
Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
Length (cm)

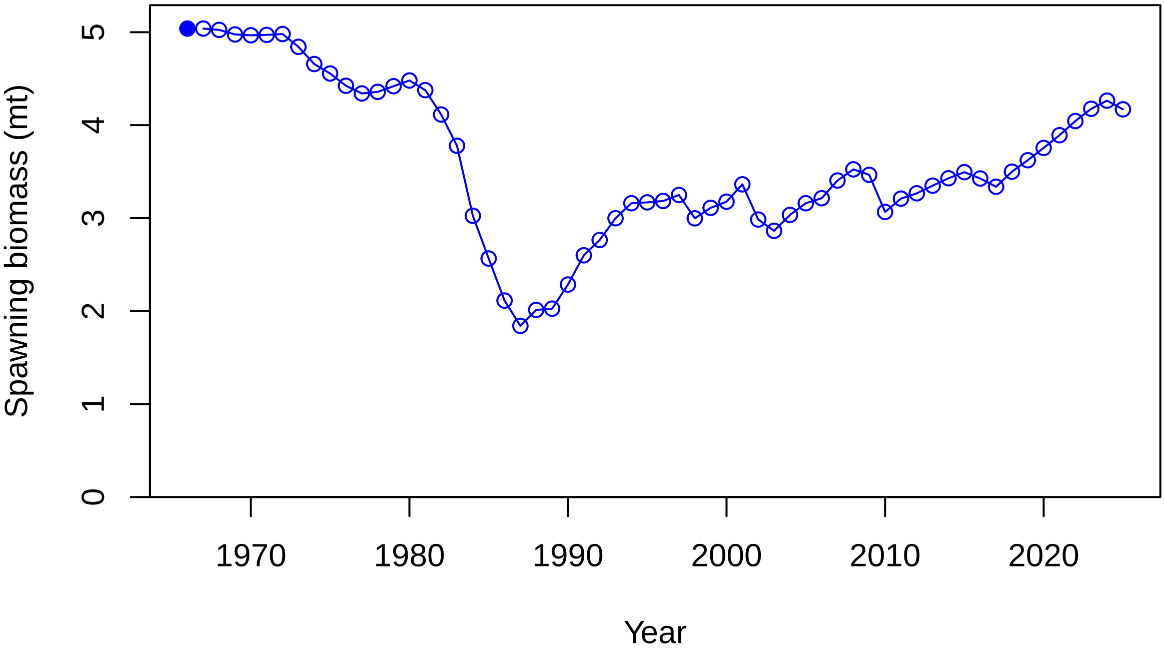


Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
Age (yr)
FISHERY
FISHERY
FISHERY

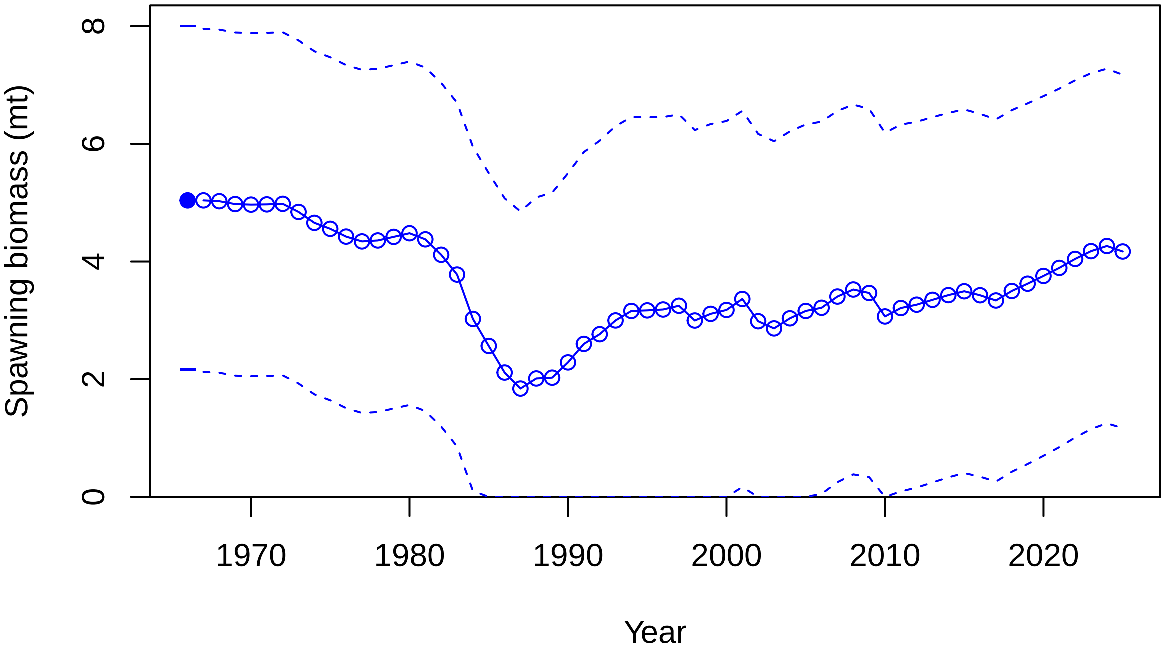


Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
Length (cm)

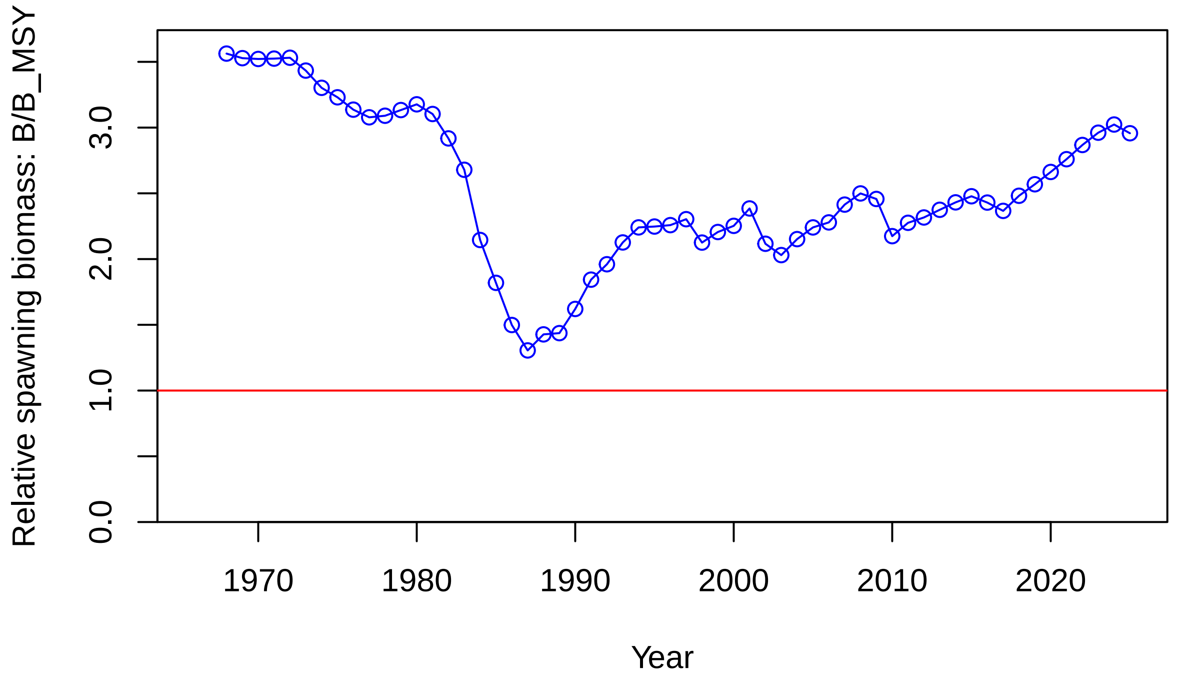

Length (cm)
Age (yr)
0
10
20
30
40
50
0
5
10
15
20
25
1
0.8
0.6
0.4
0.2

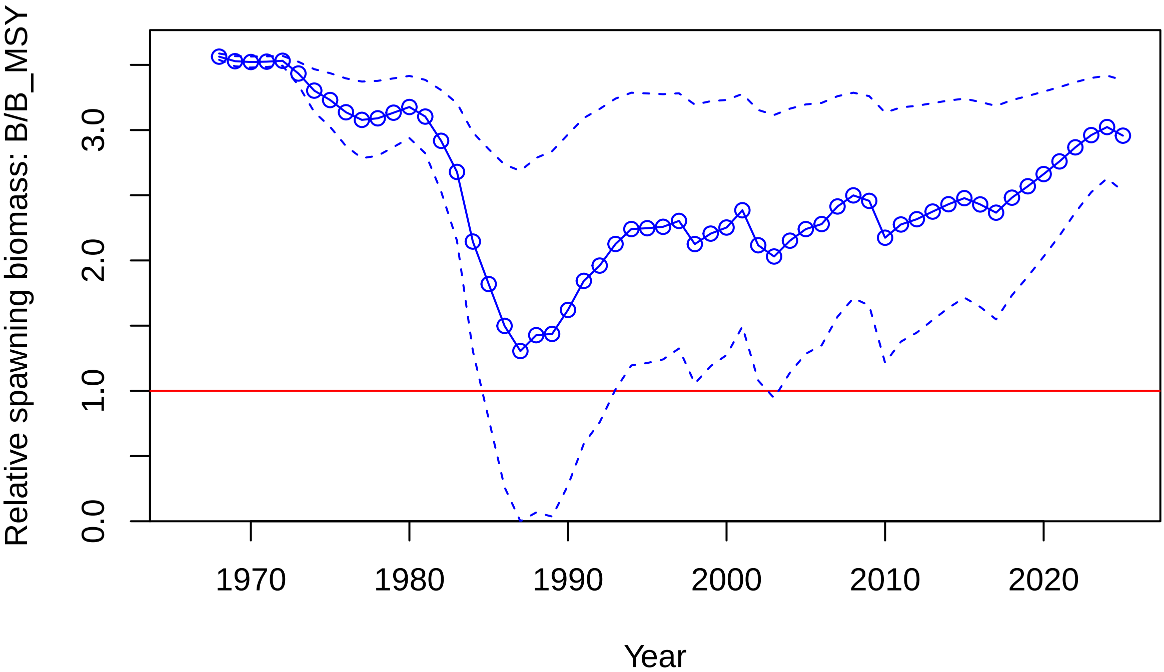


Spawning biomass (mt)
0
1
2
3
4
5
1970
1980
1990
2000
2010
2020
Year

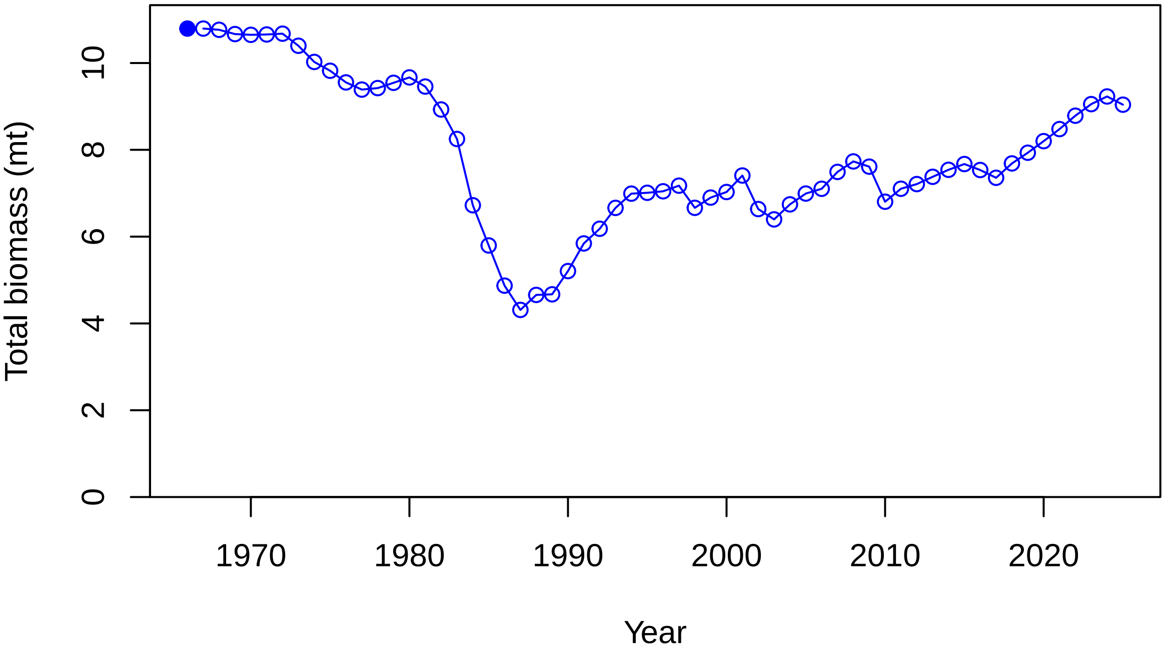


Spawning biomass (mt)
0
2
4
6
8
1970
1980
1990
2000
2010
2020
Year

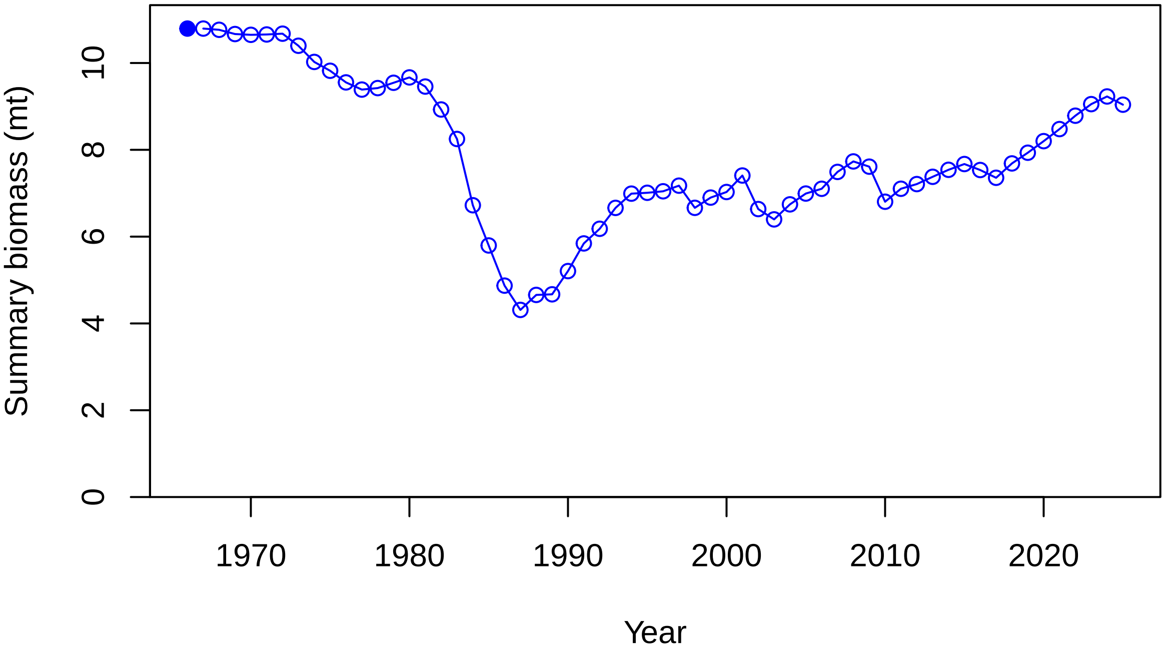

Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year



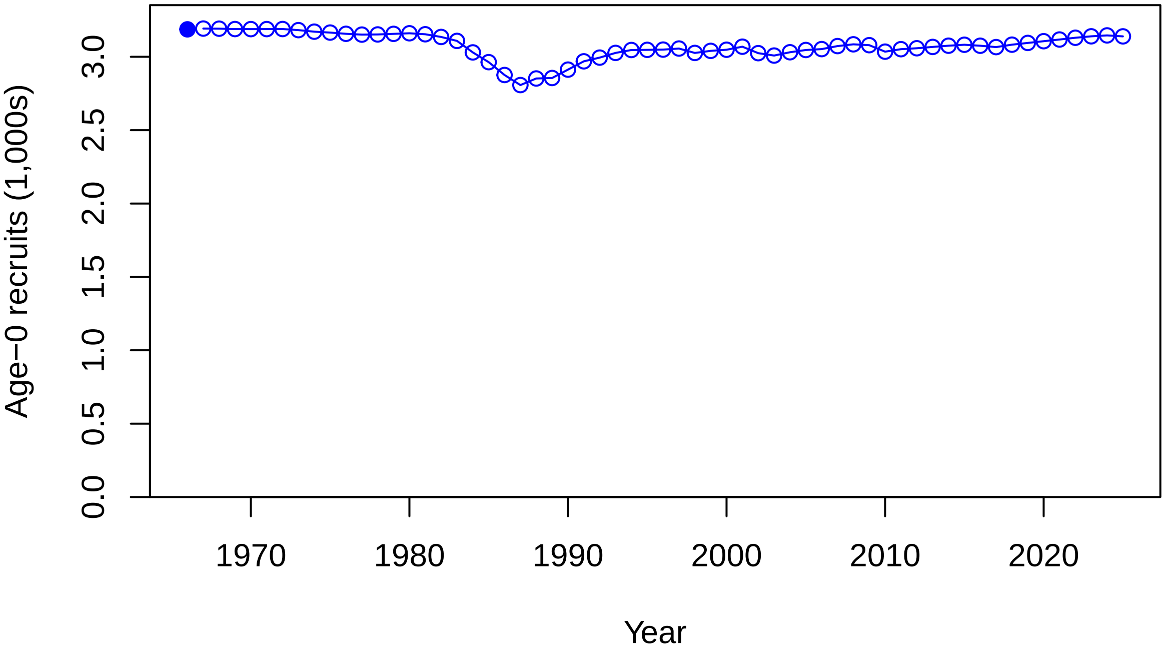
Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year



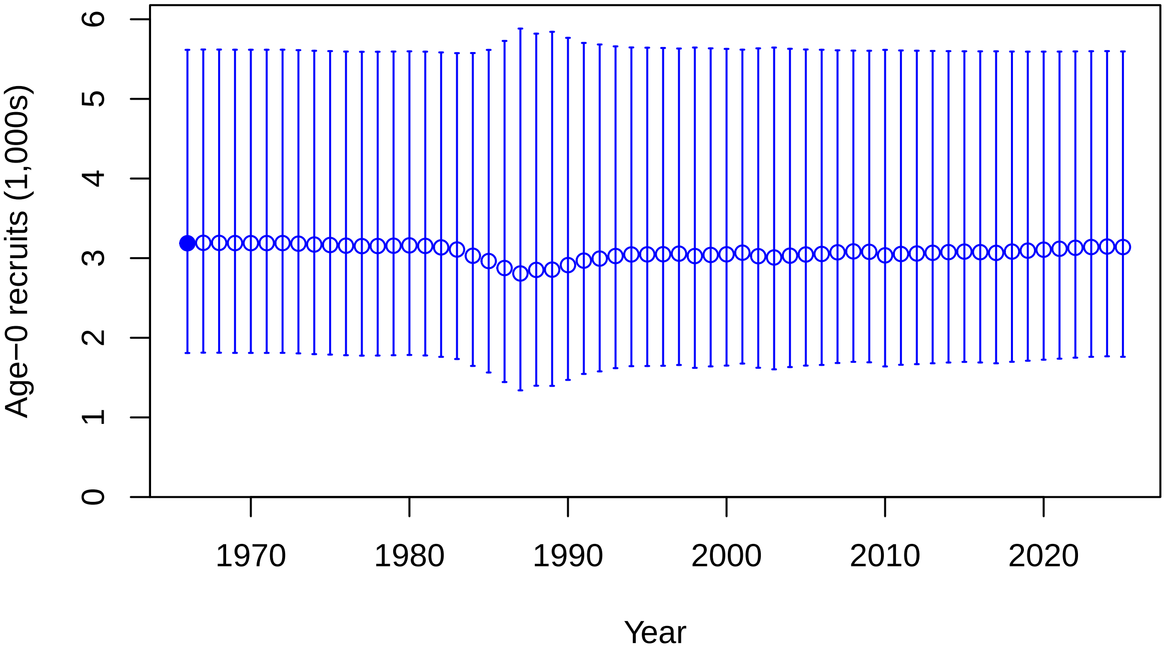
Total biomass (mt)
0
2
4
6
8
10
1970
1980
1990
2000
2010
2020
Year

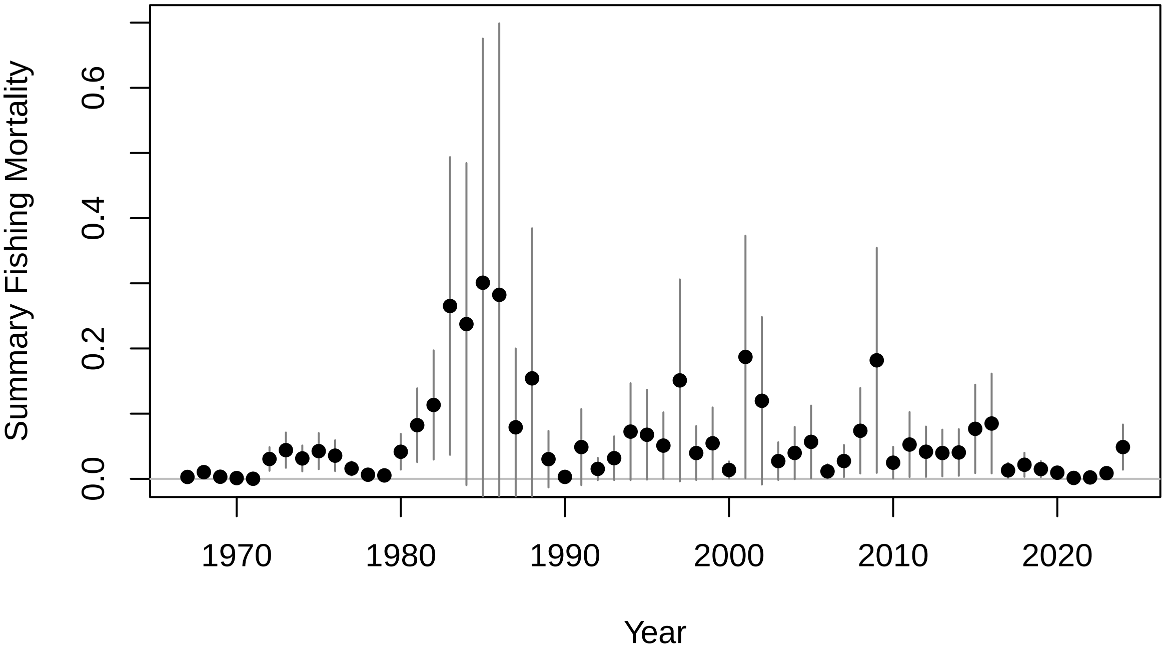


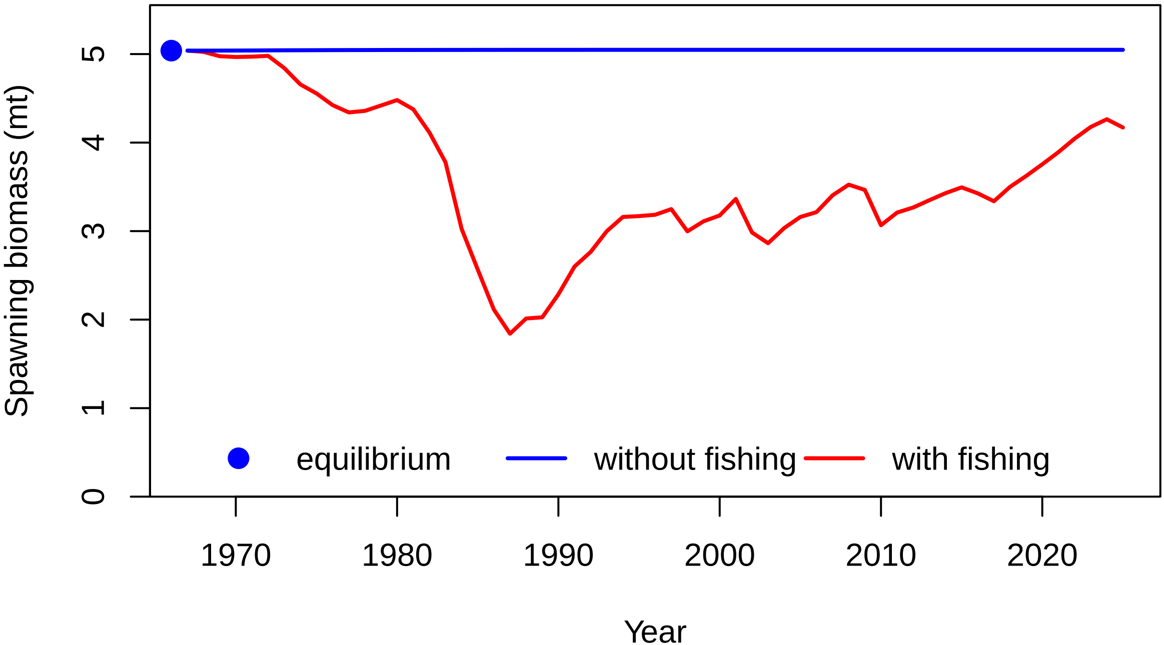
Summary biomass (mt)
0
2
4
6
8
10
1970
1980
1990
2000
2010
2020
Year



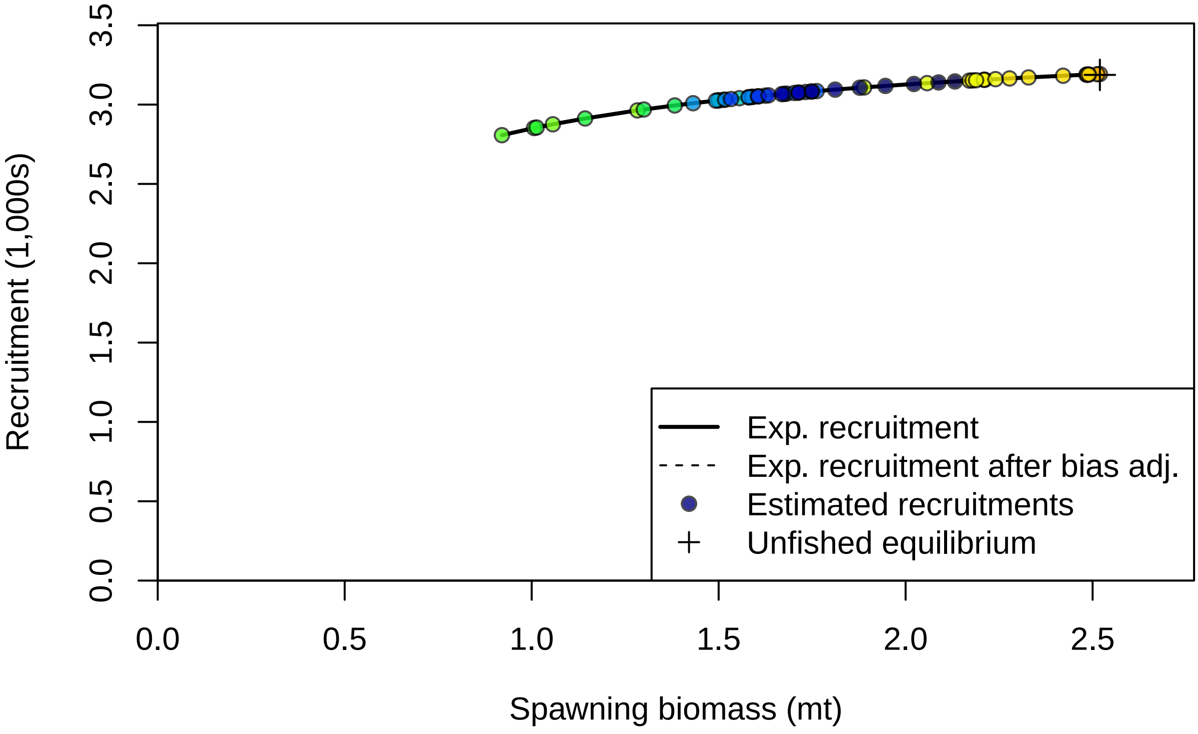
Age−0 recruits (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1970
1980
1990
2000
2010
2020
Year


Age−0 recruits (1,000s)
0
1
2
3
4
5
6
1970
1980
1990
2000
2010
2020
Year

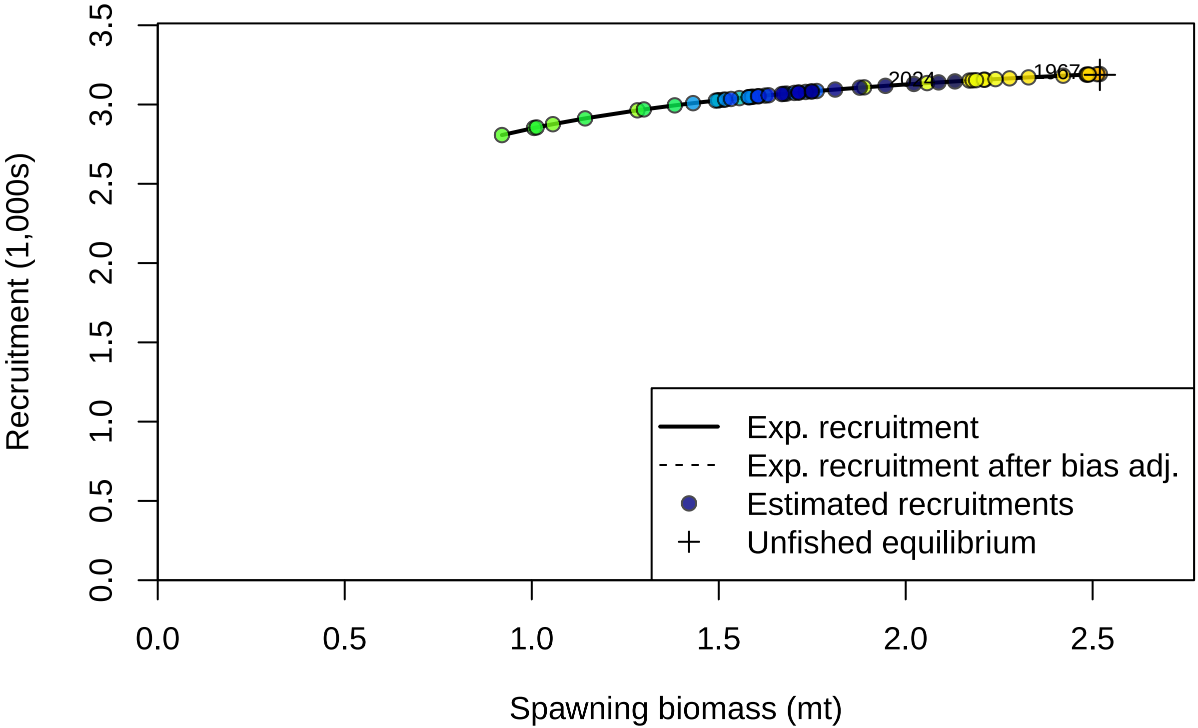

Summary Fishing Mortality
0.0
0.2
0.4
0.6
1970
1980
1990
2000
2010
2020
Year



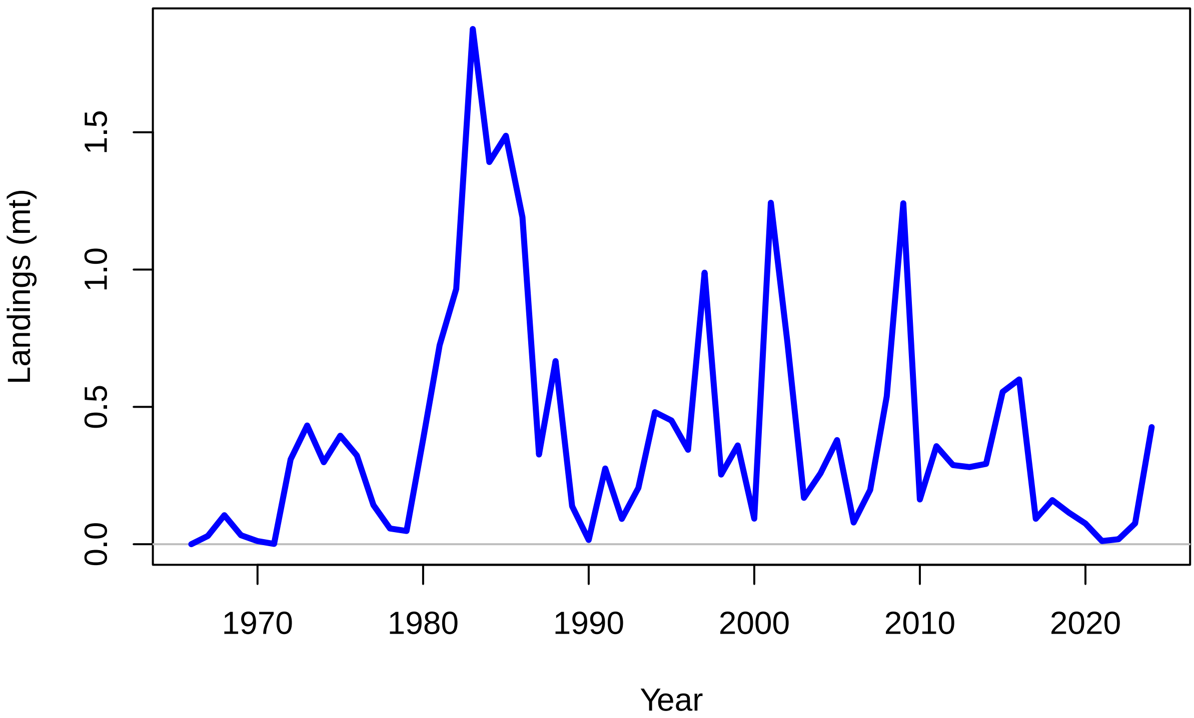
Spawning biomass (mt)
0
1
2
3
4
5
1970
1980
1990
2000
2010
2020
Year
equilibrium
without fishing
with fishing



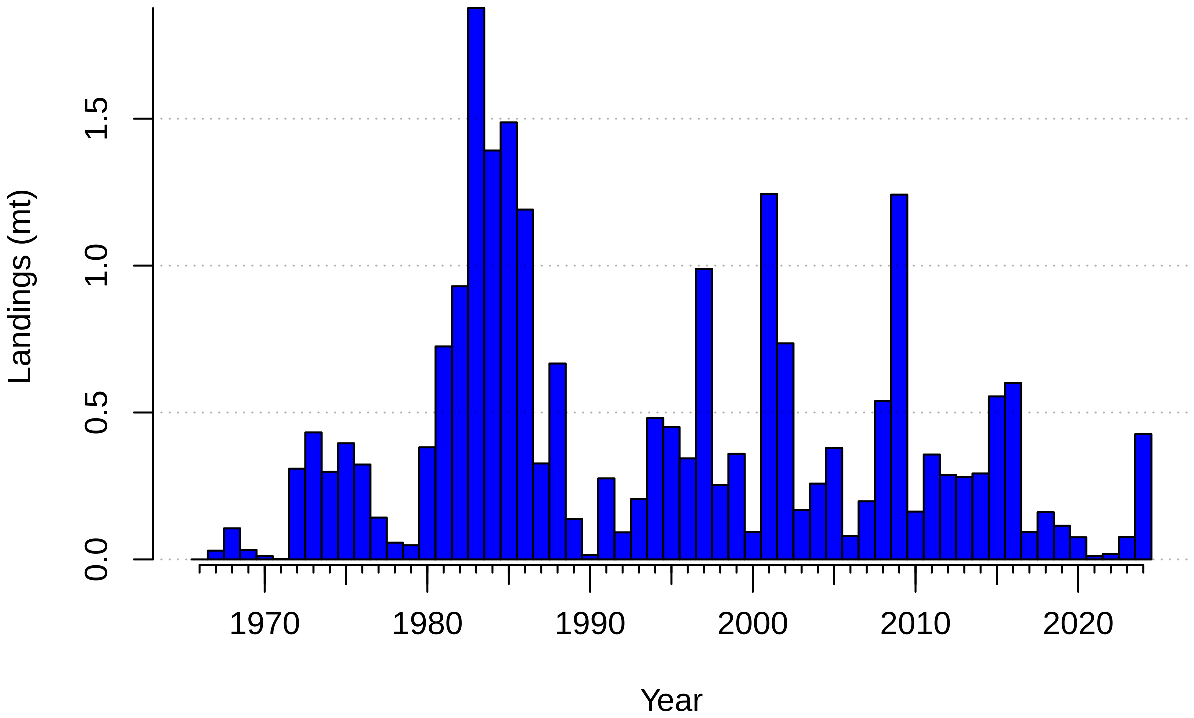
Recruitment (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0.0
0.5
1.0
1.5
2.0
2.5
Spawning biomass (mt)
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium

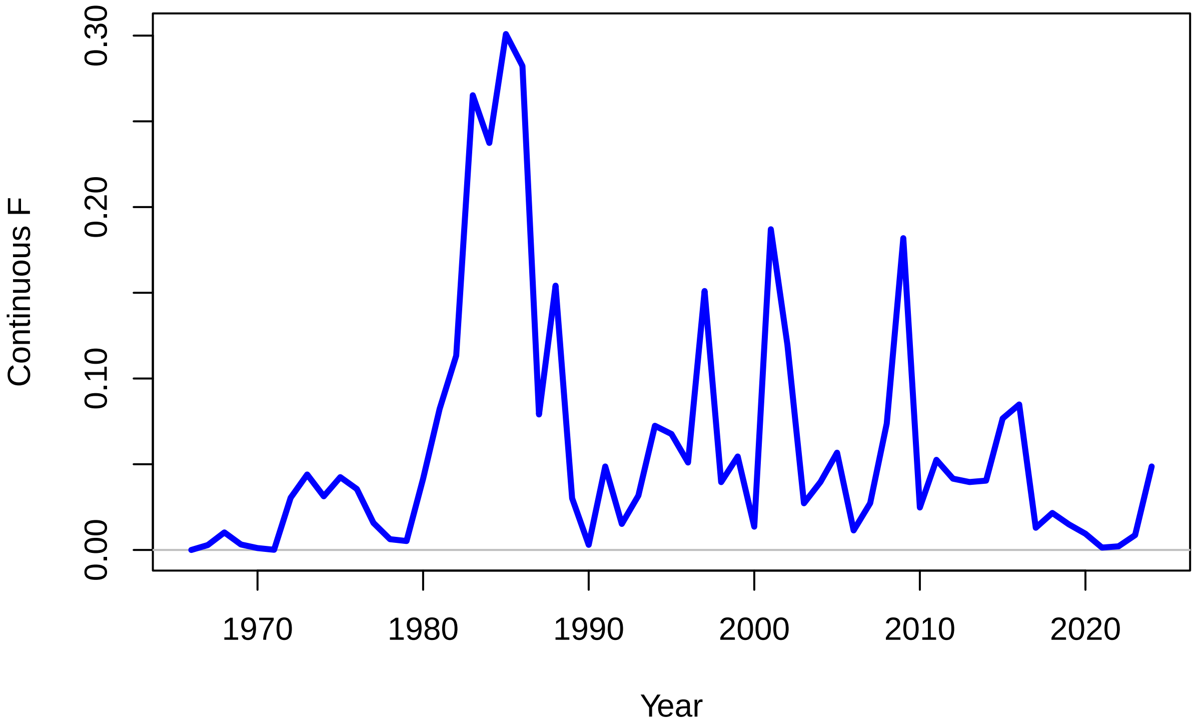


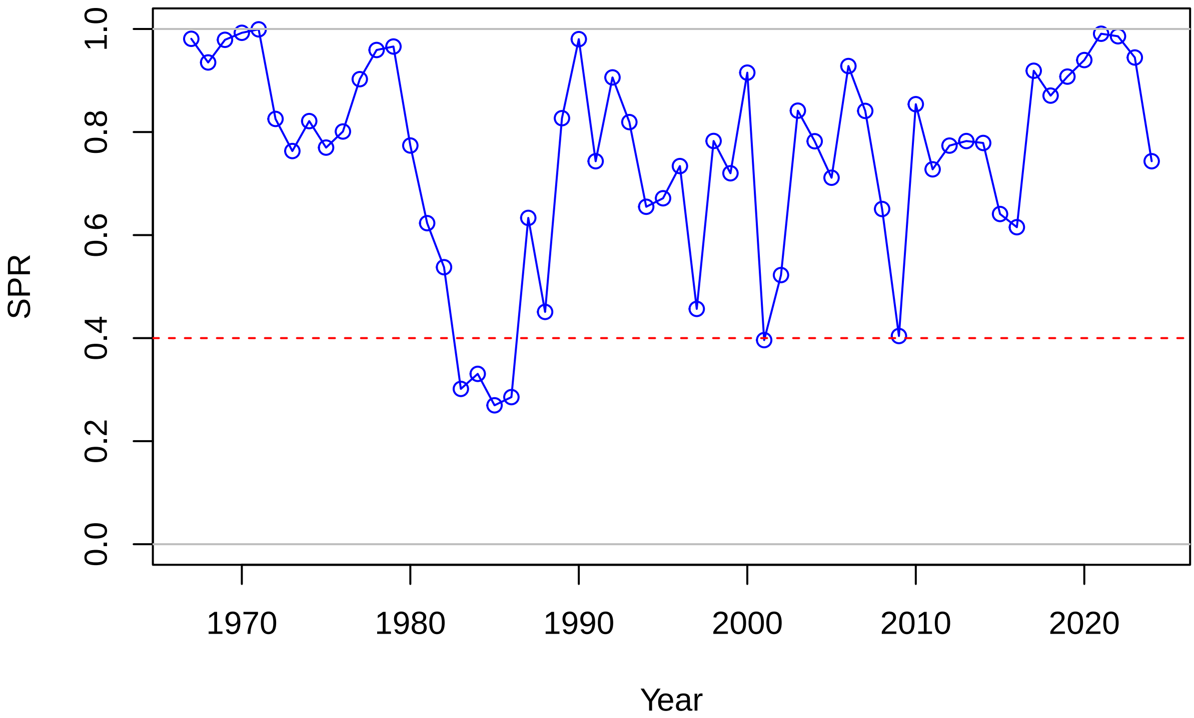
Recruitment (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Spawning biomass (mt)
0.0
0.5
1.0
1.5
2.0
2.5
2024
1967
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium



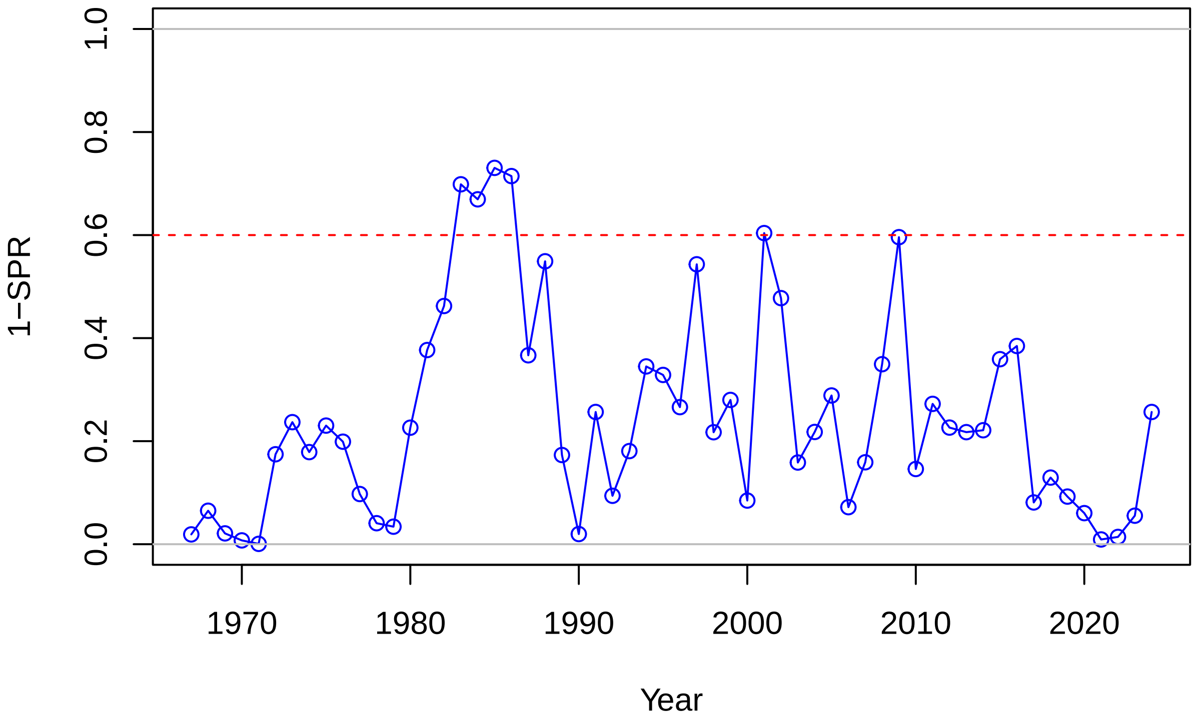
Landings (mt)
0.0
0.5
1.0
1.5
1970
1980
1990
2000
2010
2020
Year

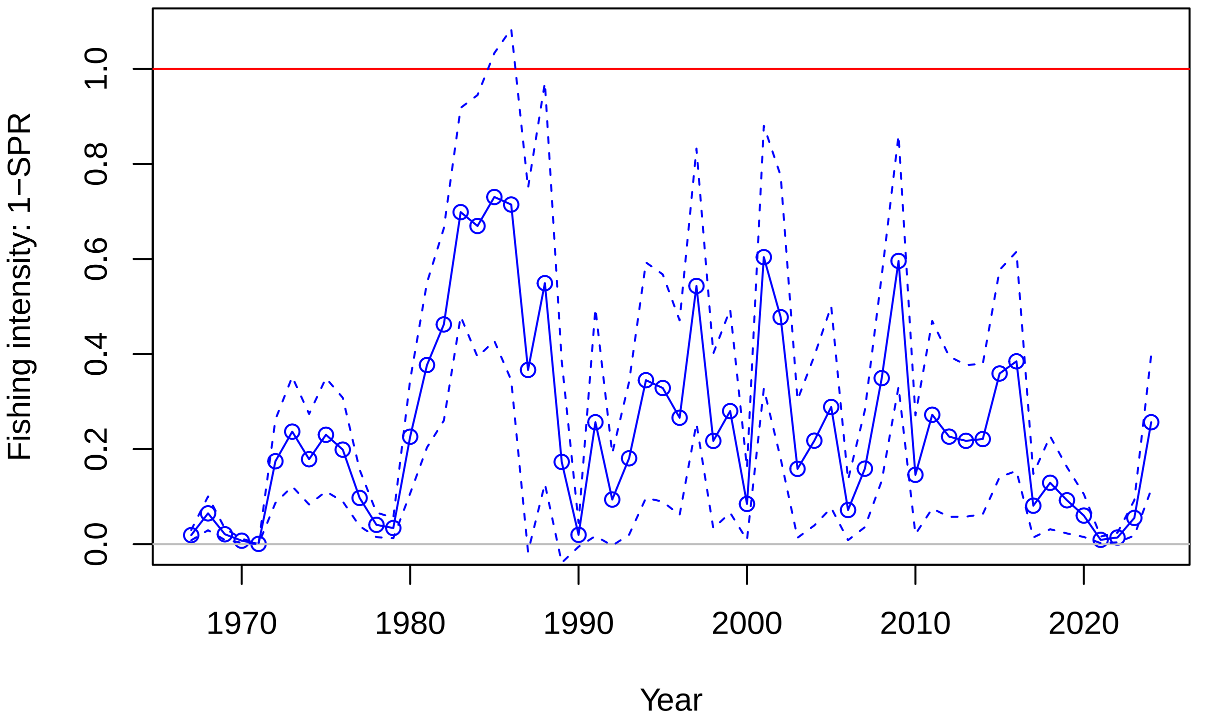

Landings (mt)
0.0
0.5
1.0
1.5
1970
1980
1990
2000
2010
2020
Year

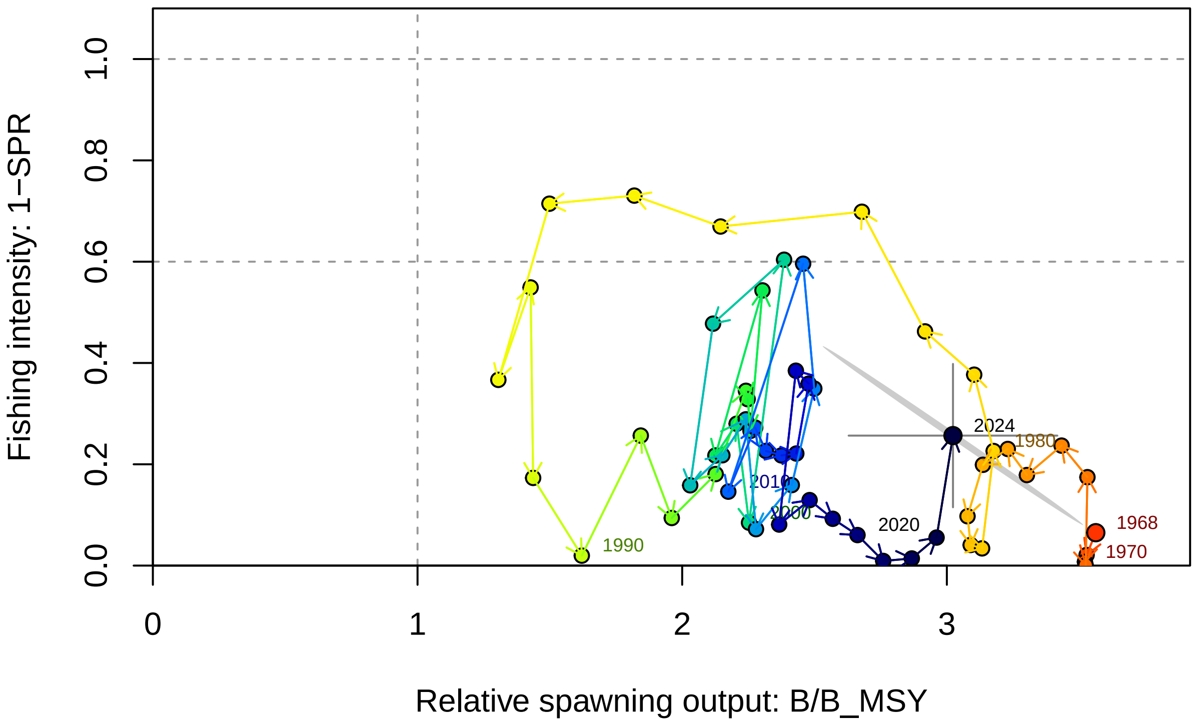

Continuous F
0.00
0.10
0.20
0.30
1970
1980
1990
2000
2010
2020
Year



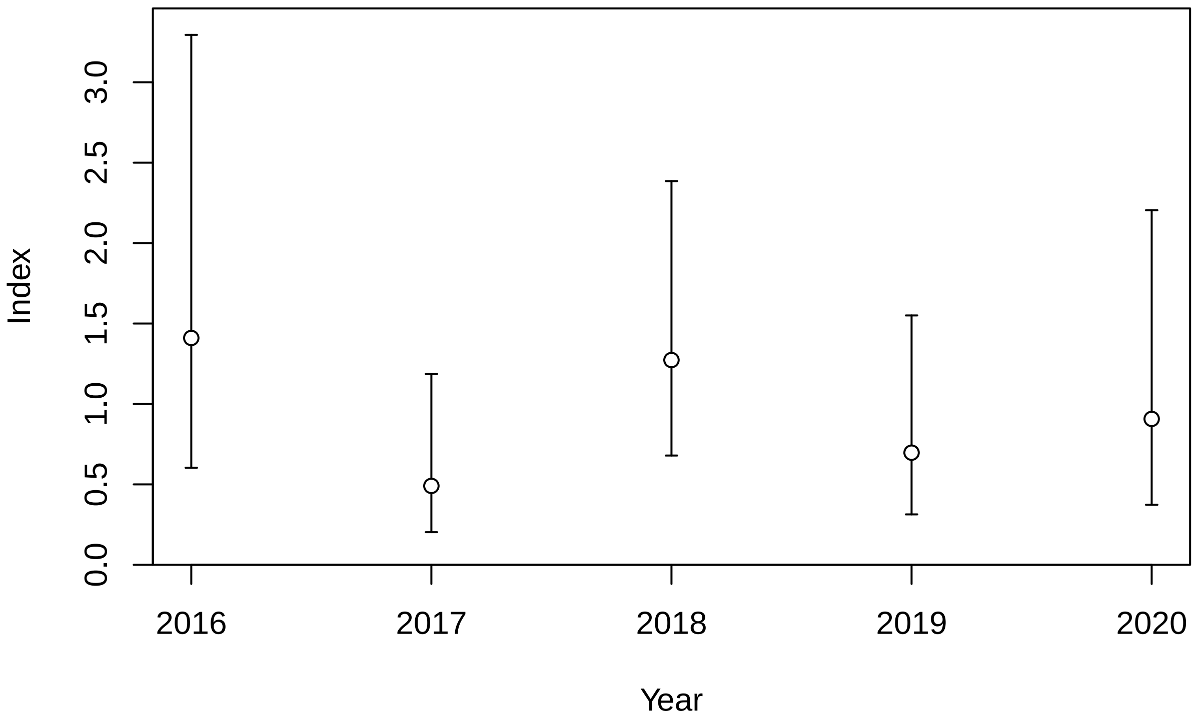
SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year

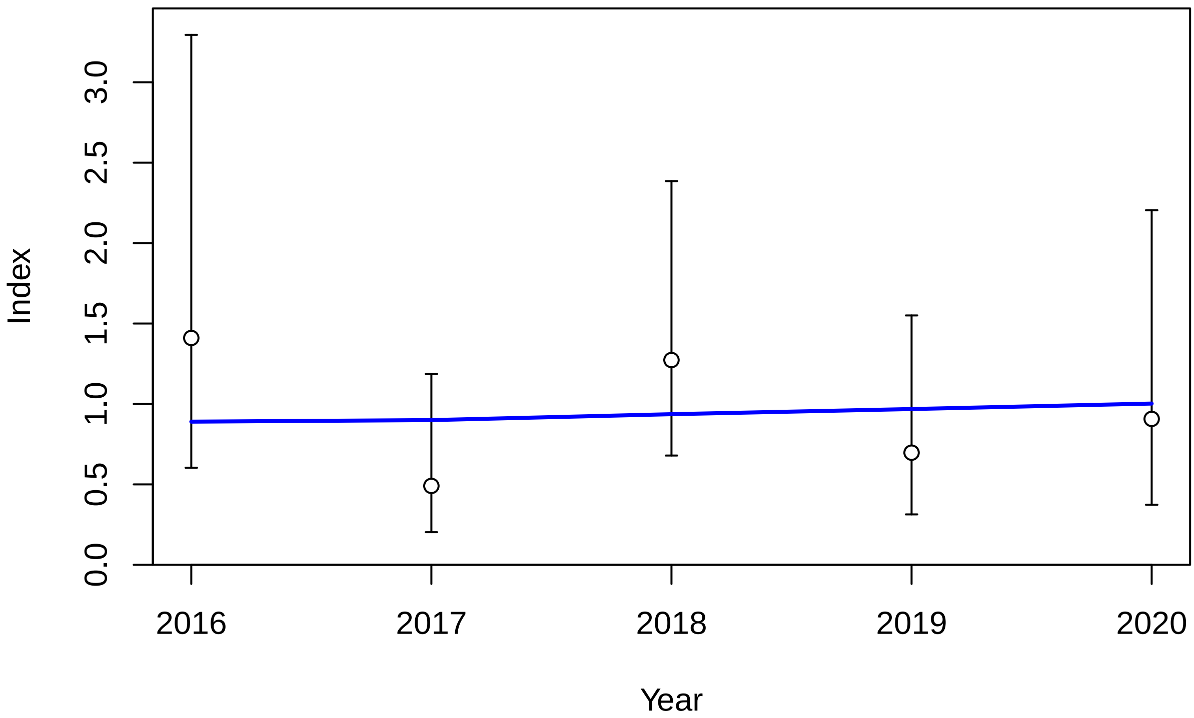

1−SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year

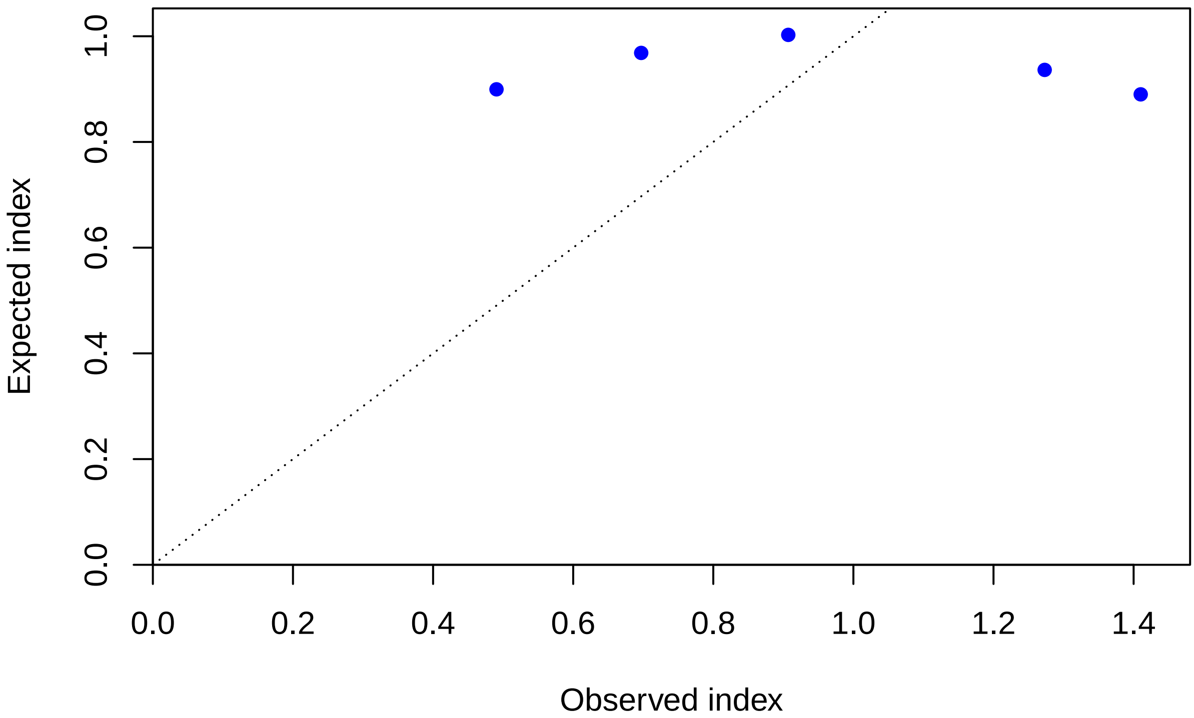

Fishing intensity: 1−SPR
1.0
0.8
0.6
0.4
0.2
0.0
1970
1980
1990
2000
2010
2020
Year



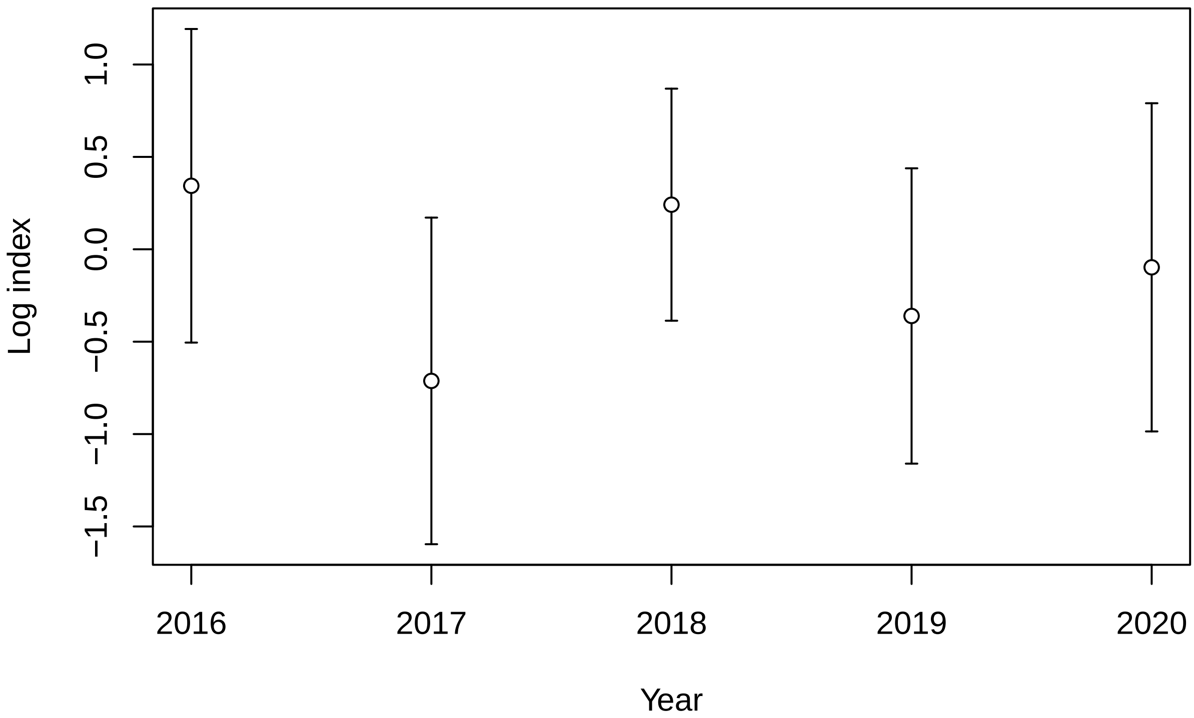
Fishing intensity: 1−SPR
Relative spawning output: B/B_MSY
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
1968
1970
1980
1990
2000
2010
2020
2024


Index
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2016
2017
2018
2019
2020
Year

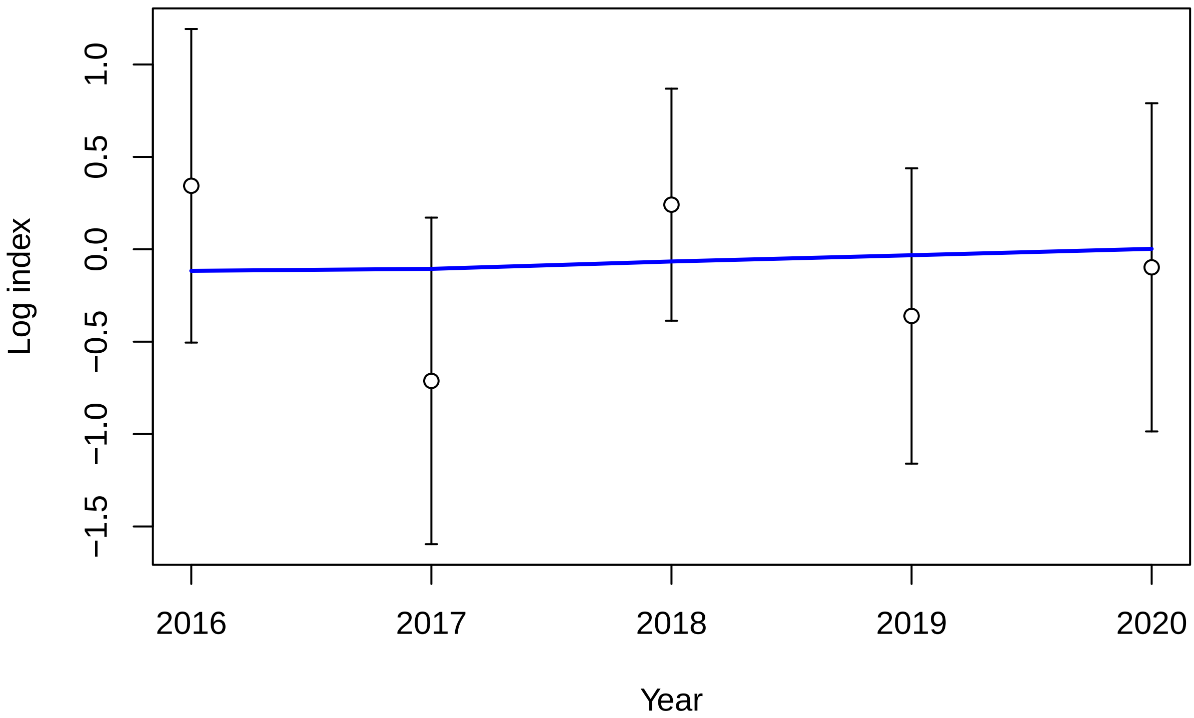

Index
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2016
2017
2018
2019
2020
Year

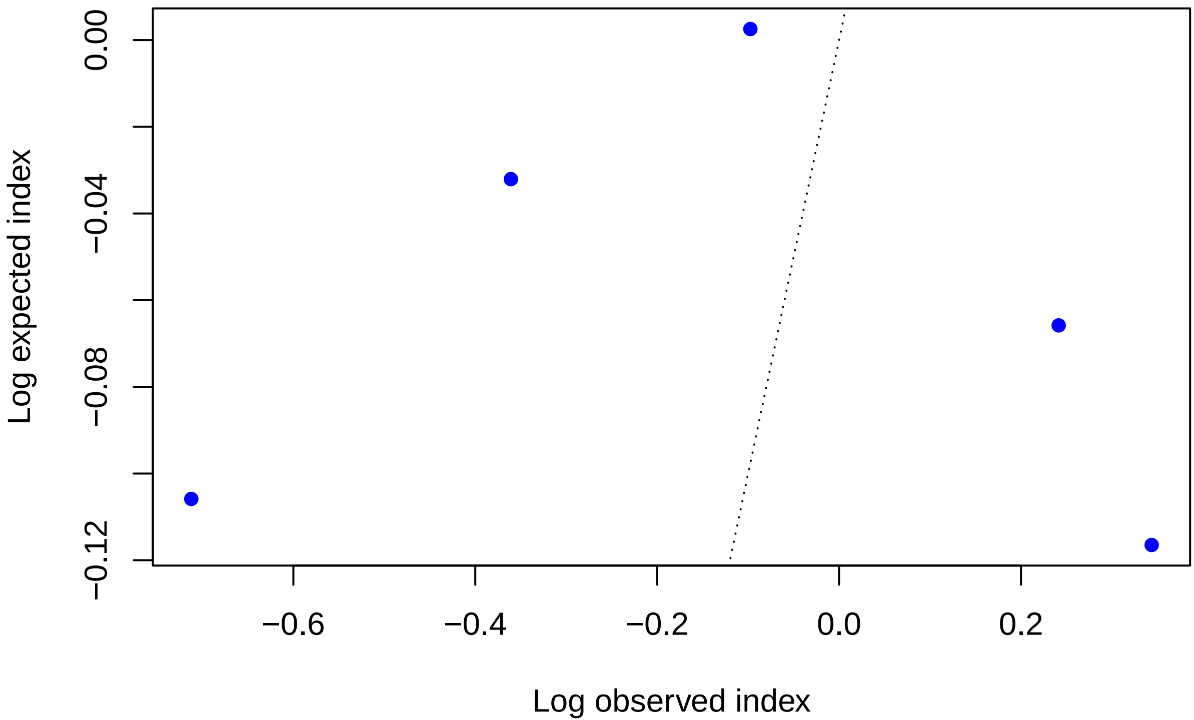


Expected index
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Observed index

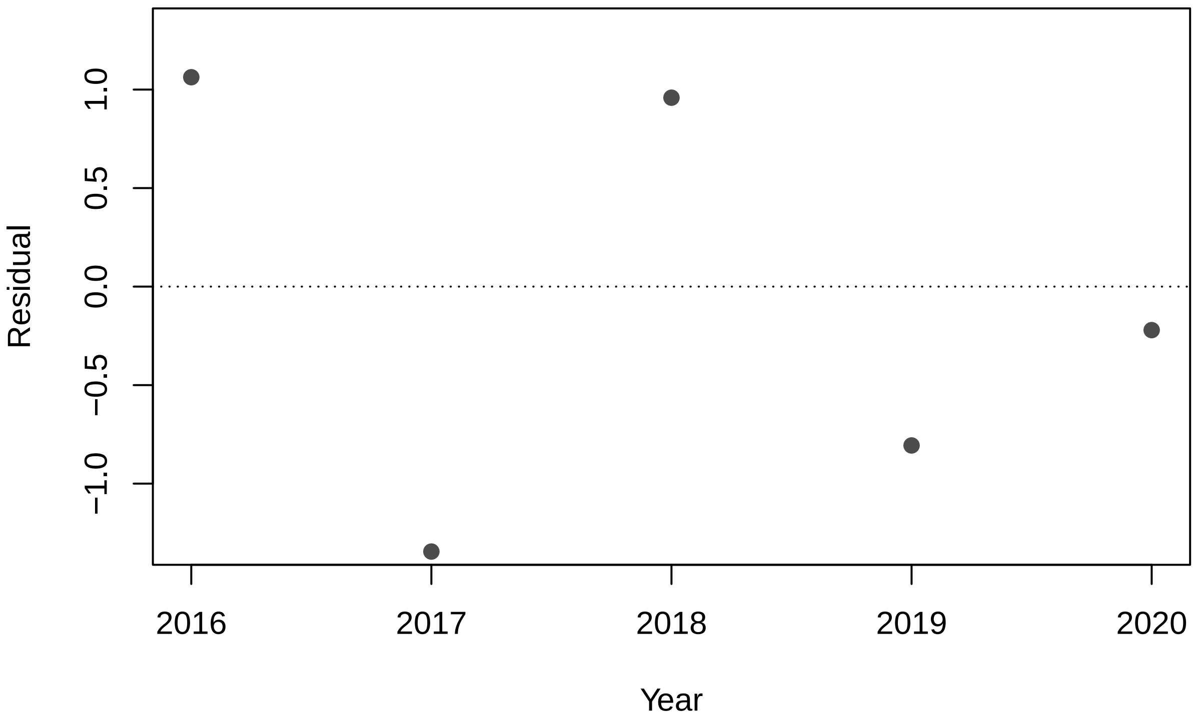


Log index
1.0
0.5
0.0
−0.5
−1.0
−1.5
2016
2017
2018
2019
2020
Year

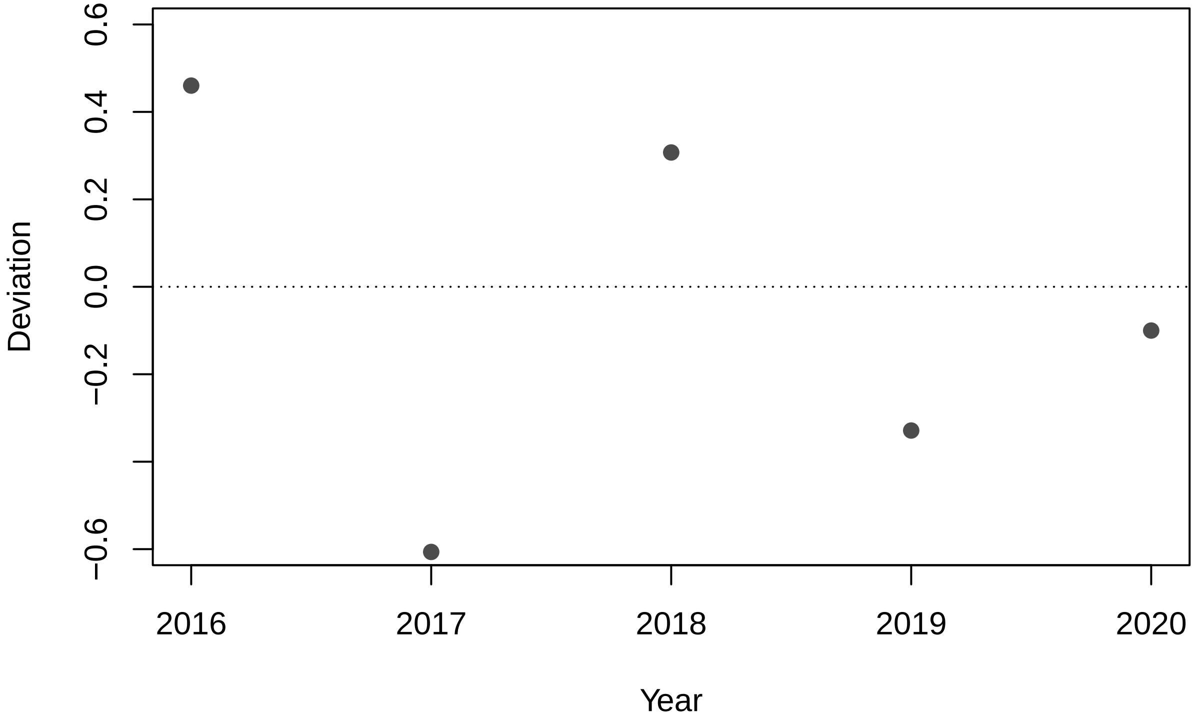

Log index
1.0
0.5
0.0
−0.5
−1.0
−1.5
2016
2017
2018
2019
2020
Year

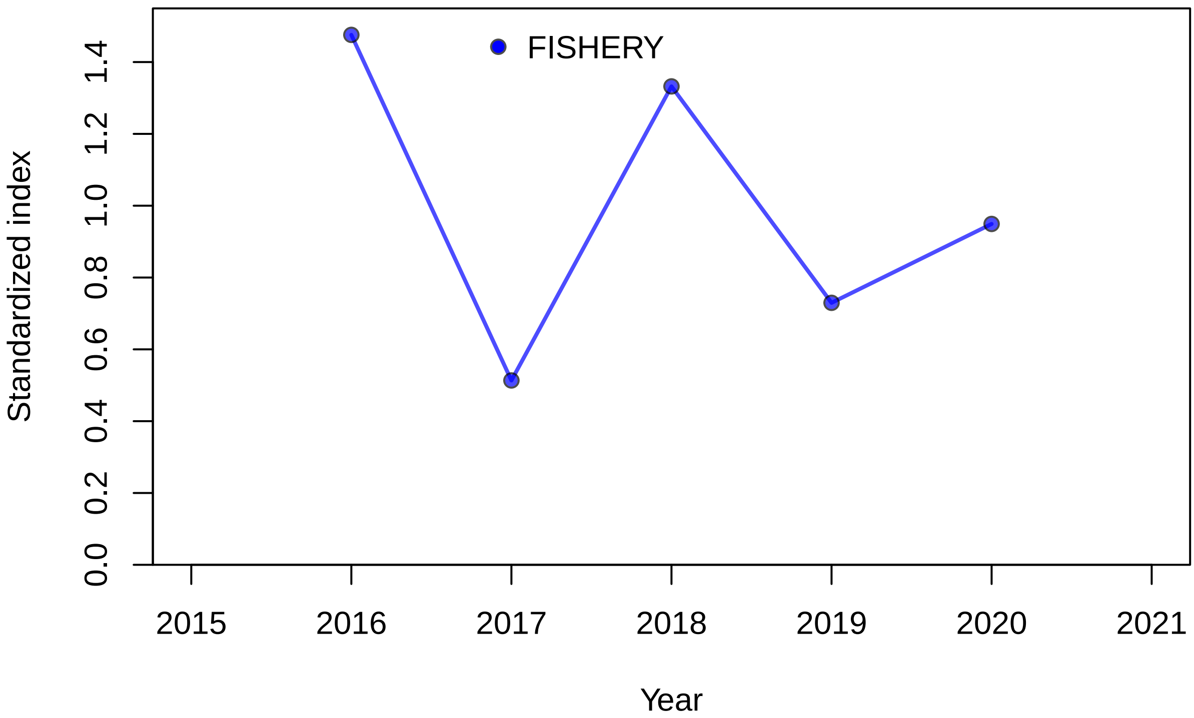


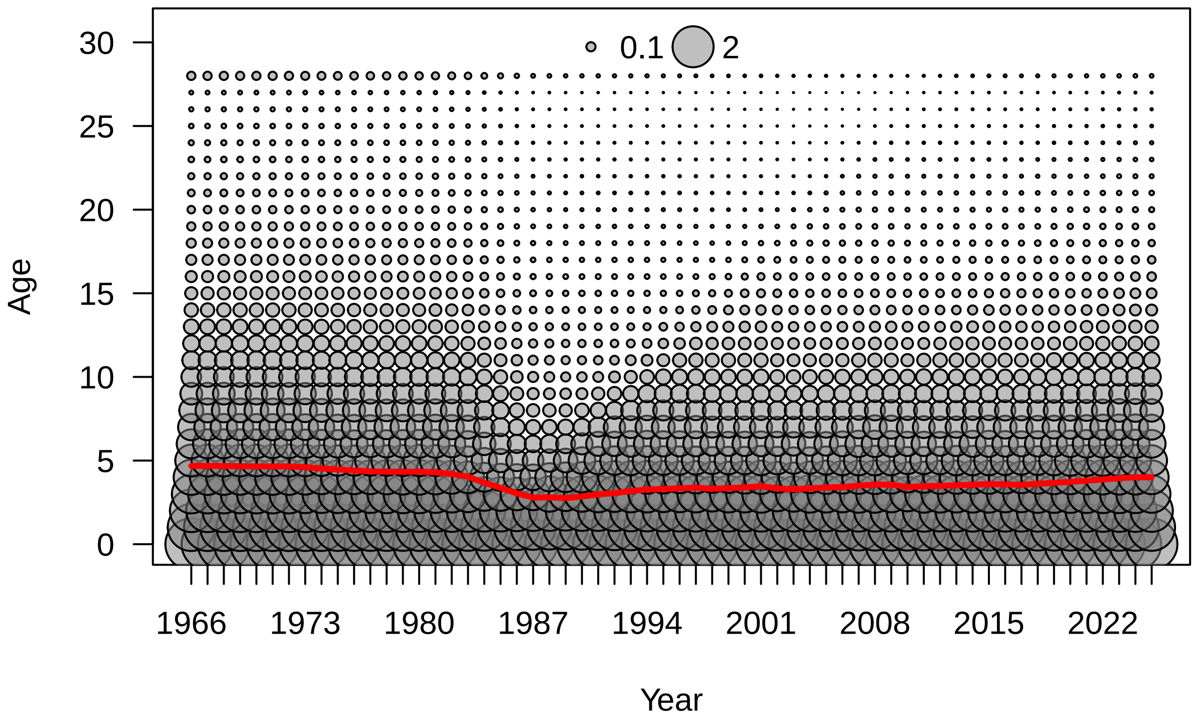
0.00
−0.04
−0.08
−0.12
Log expected index
−0.6
−0.4
−0.2
0.0
0.2
Log observed index



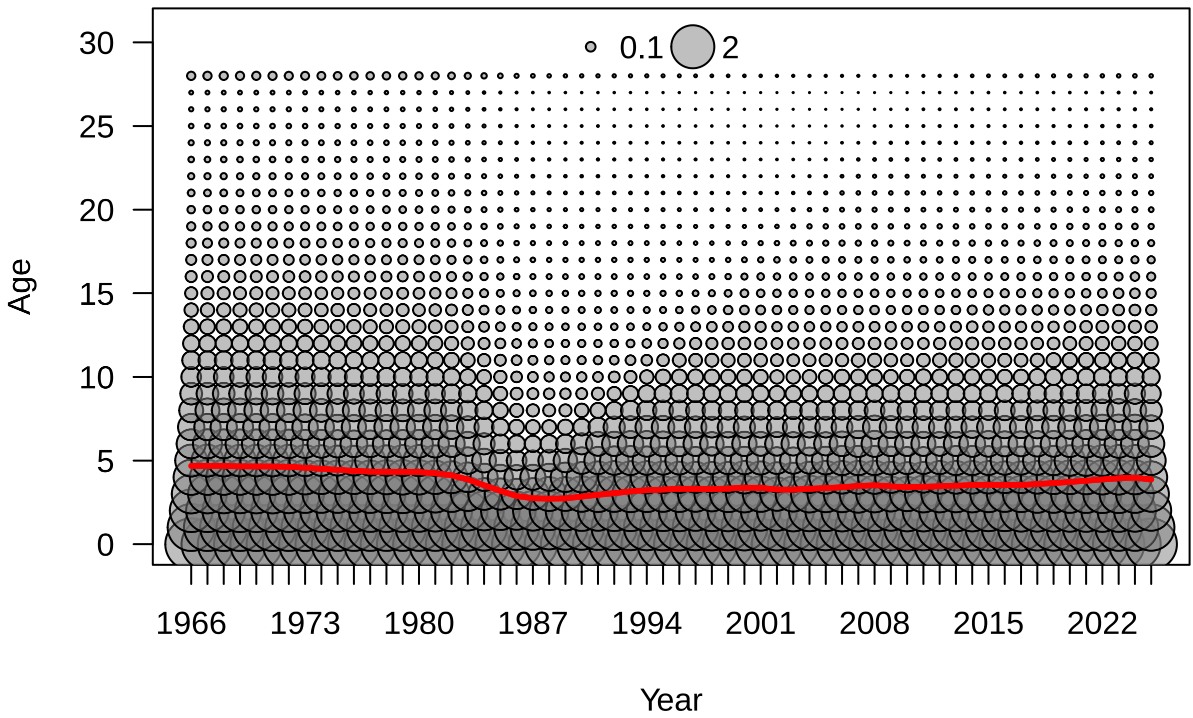
Residual
1.0
0.5
0.0
−0.5
−1.0
2016
2017
2018
2019
2020
Year


0.6
0.4
0.2
0.0
−0.2
−0.6
Deviation
2016
2017
2018
2019
2020
Year

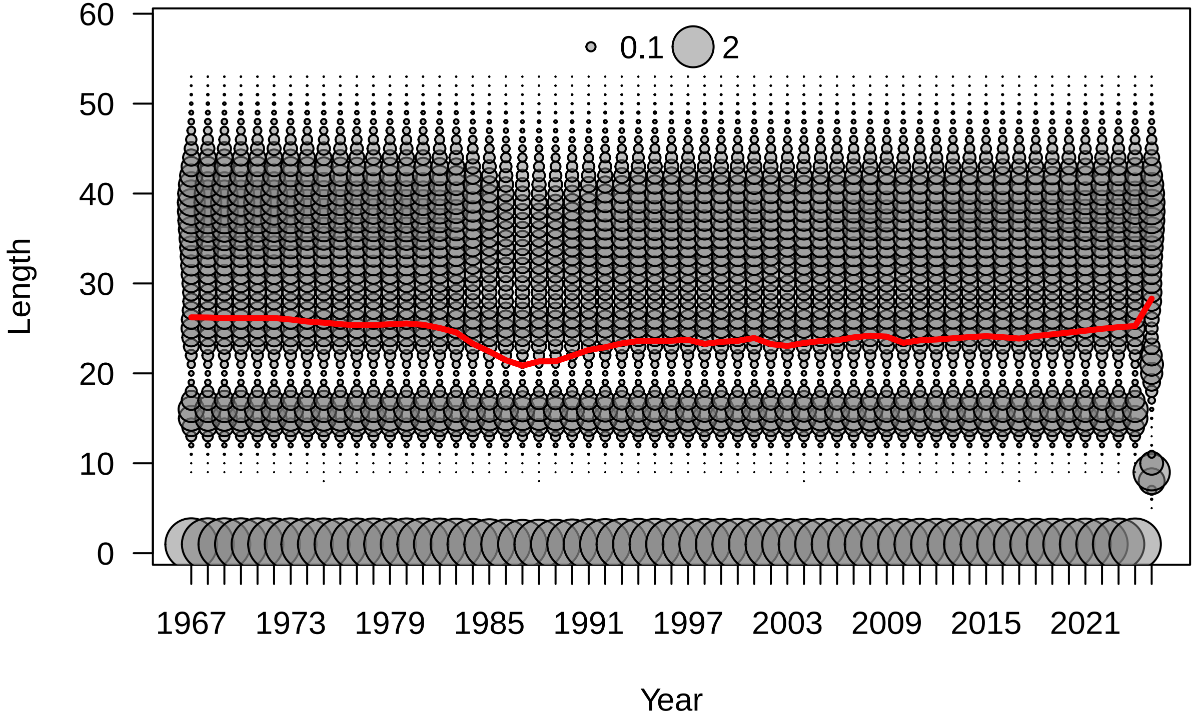

FISHERY
Standardized index
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
2015
2016
2017
2018
2019
2020
2021
Year



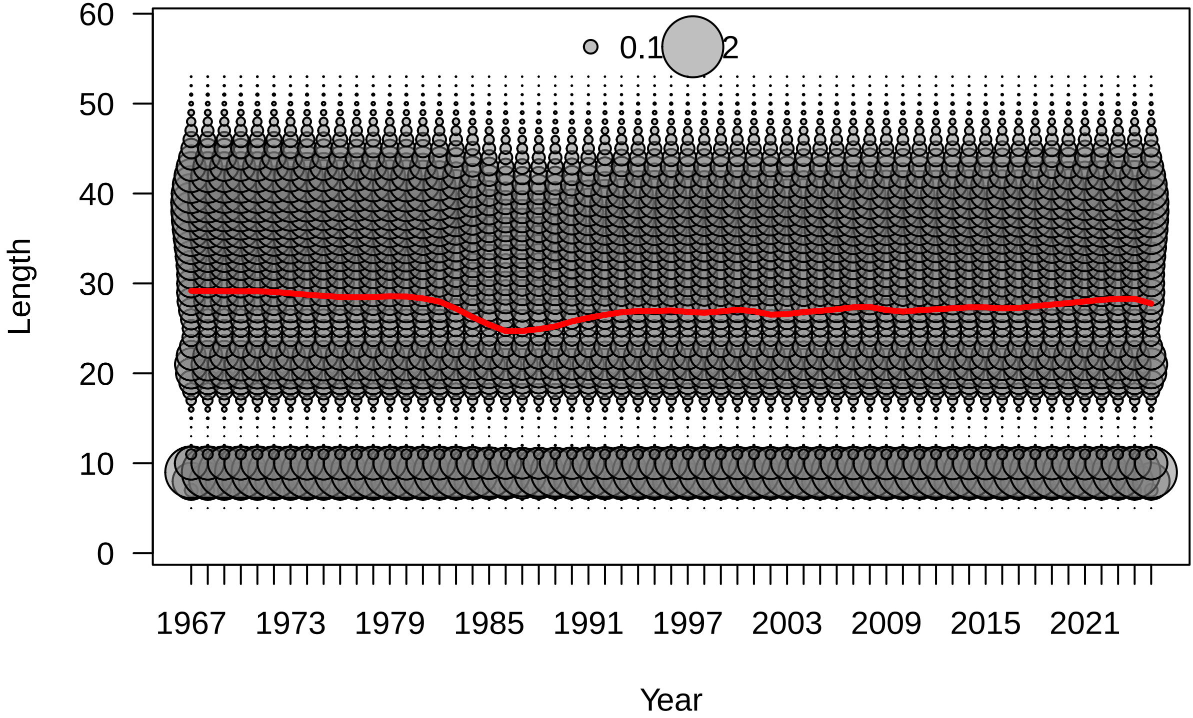
0.1
2
Age
30
25
20
15
10
5
0
Year
1966
1973
1980
1987
1994
2001
2008
2015
2022



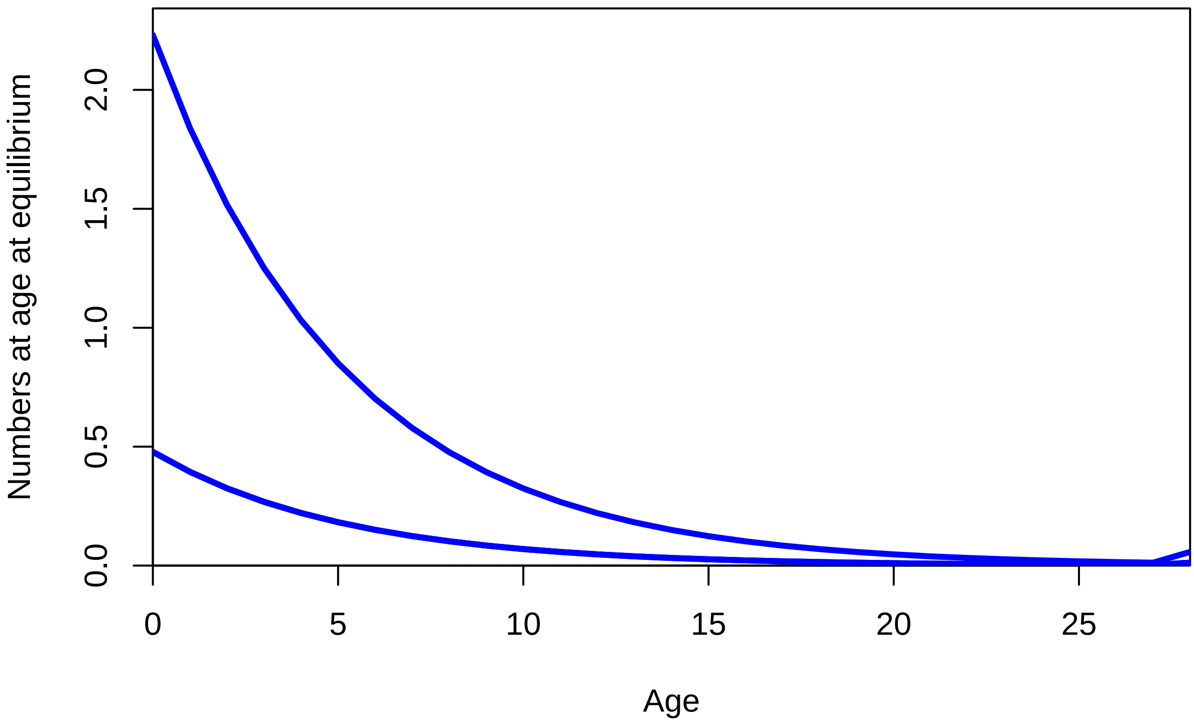
0.1
2
Age
30
25
20
15
10
5
0
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year



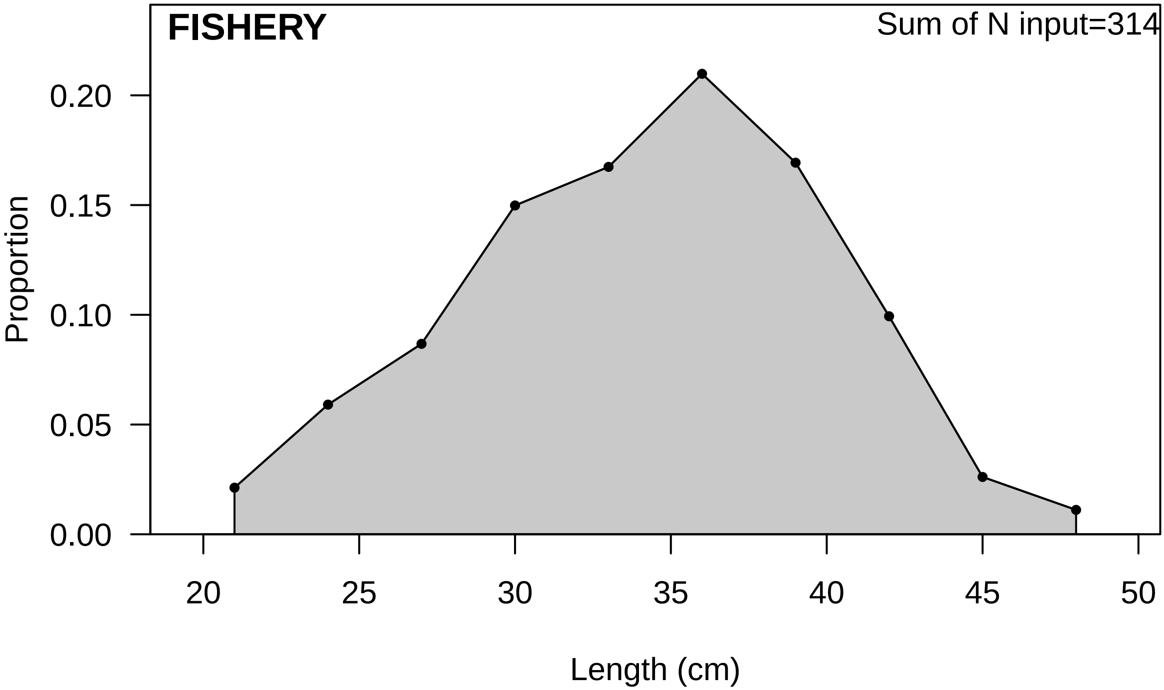
0.1
2
60
50
40
30
20
10
0
Length
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year

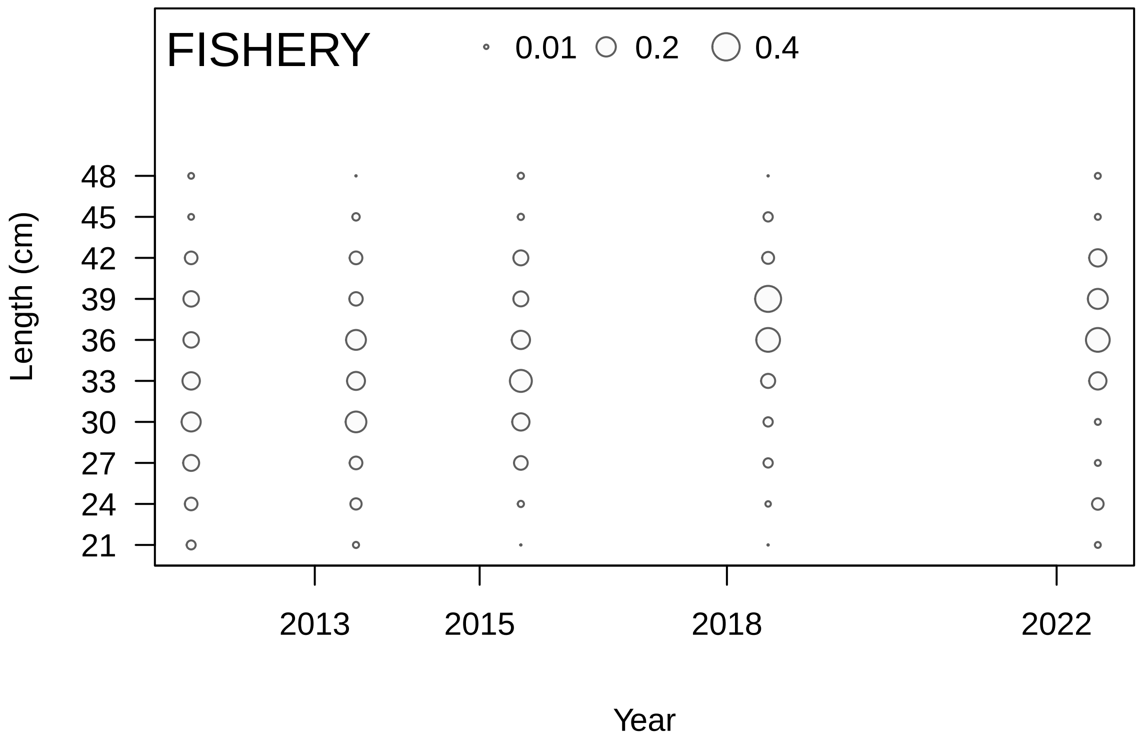


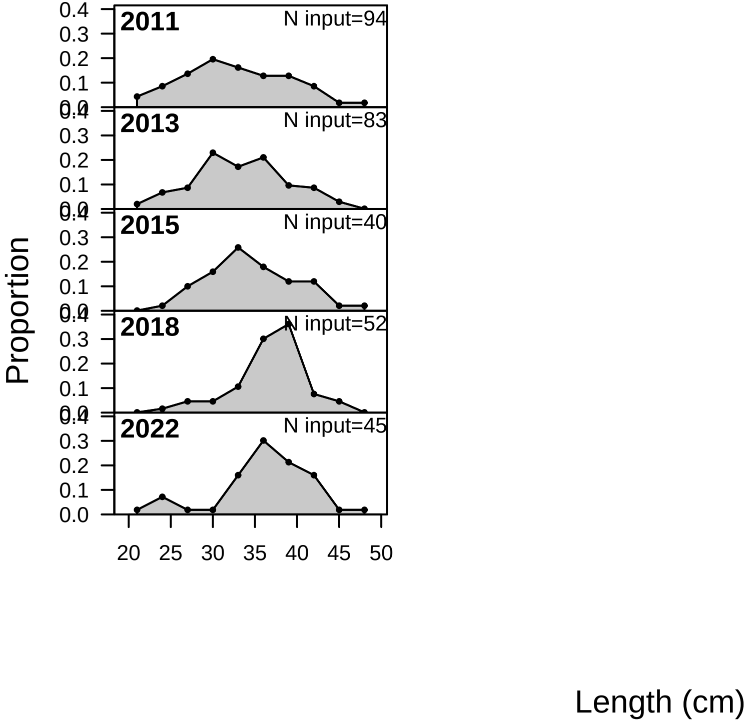
0.1
2
Length
60
50
40
30
20
10
0
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year



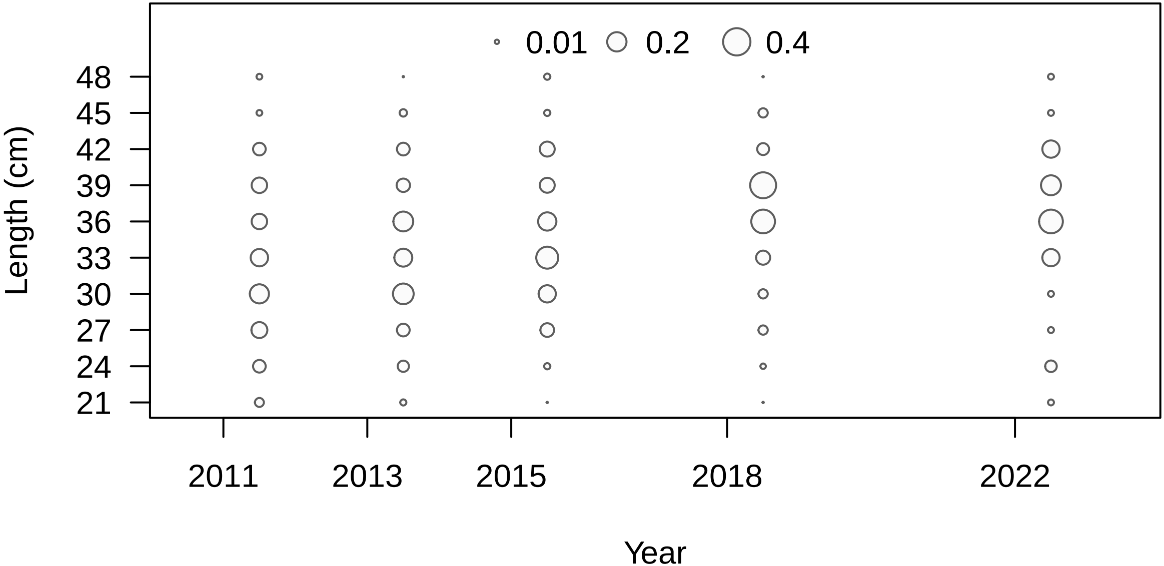
Numbers at age at equilibrium
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20
25
Age

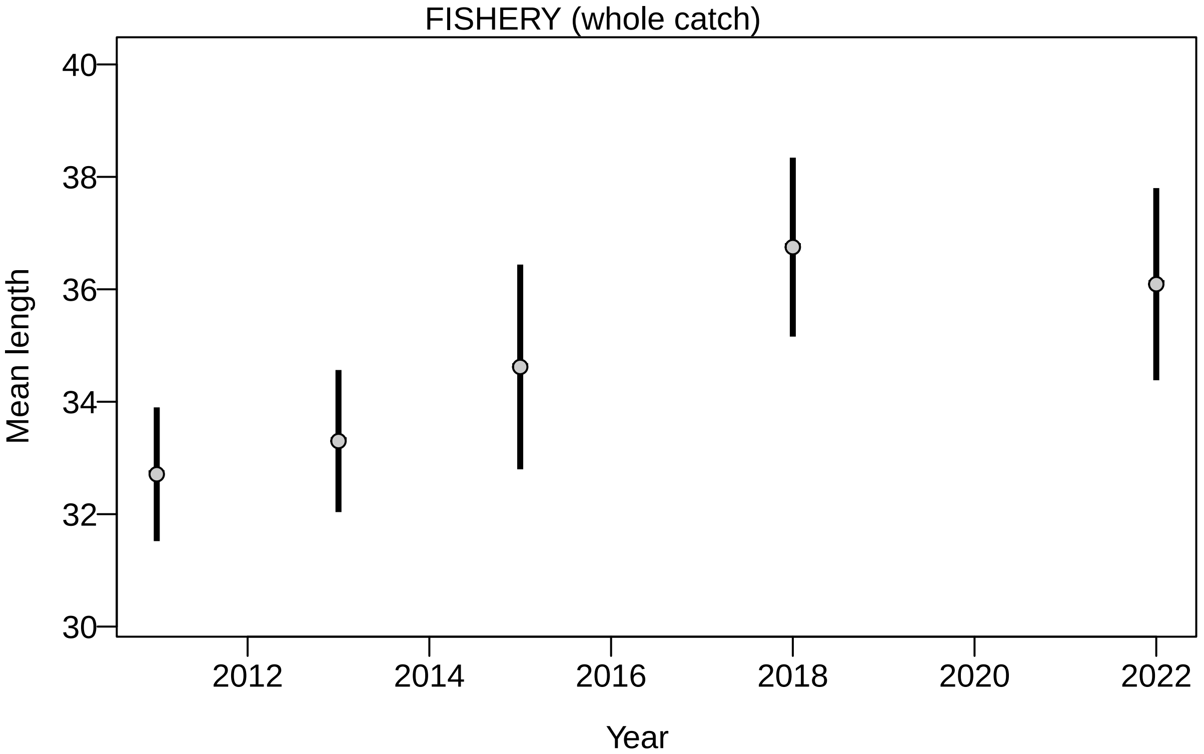

FISHERY
Sum of N input=314
Proportion
0.00
0.05
0.10
0.15
0.20
20
25
30
35
40
45
50
Length (cm)

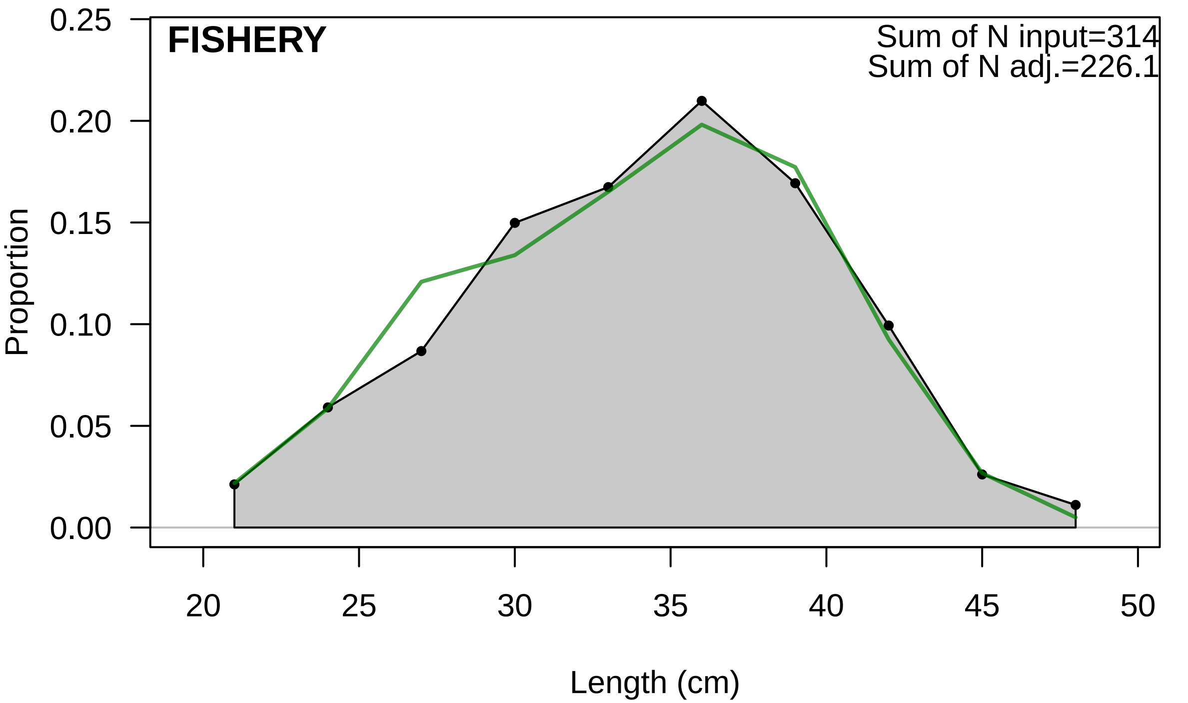

FISHERY
0.01
0.2
0.4
Length (cm)
48
45
42
39
36
33
30
27
24
21
2013
2015
2018
2022
Year

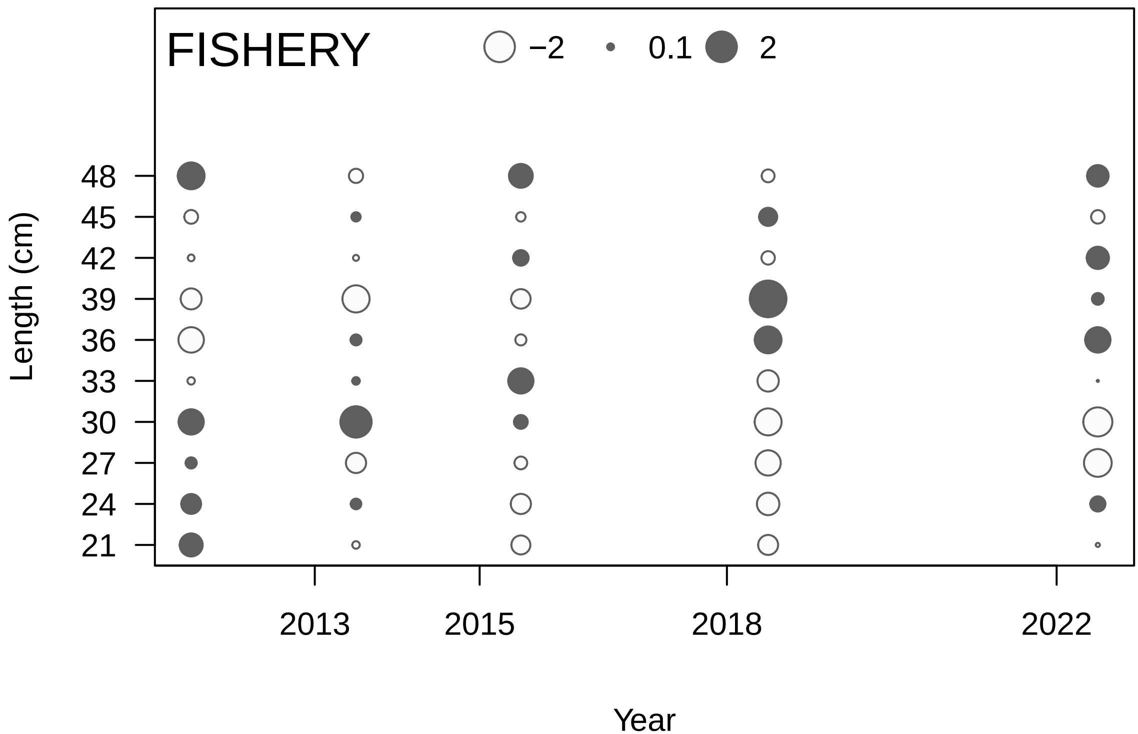


Proportion
2011
N input=94
2013
N input=83
2015
N input=40
2018
N input=52
2022
N input=45
0.0
0.1
0.2
0.3
0.4
20
25
30
35
40
45
50
Length (cm)

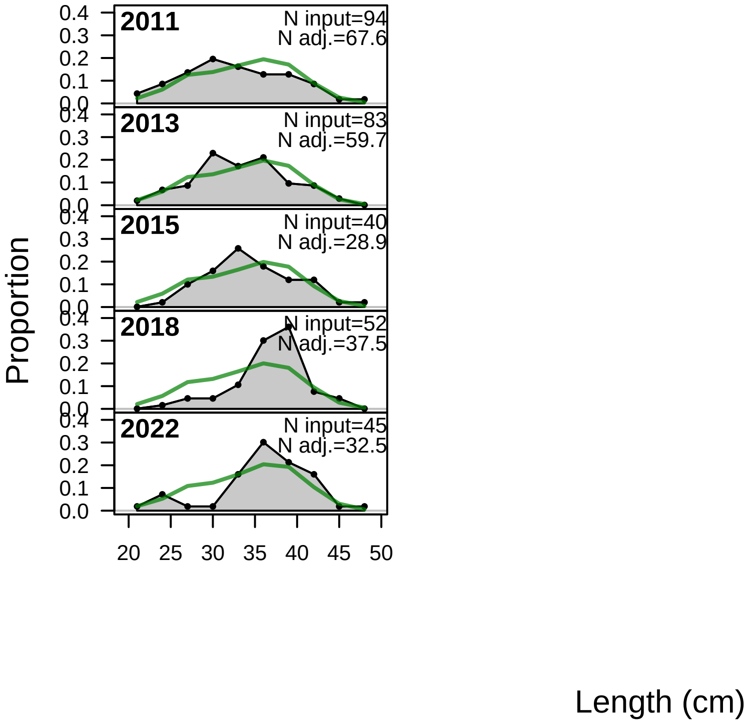

0.01
0.2
0.4
48
45
42
39
36
33
30
27
24
21
Length (cm)
2011
2013
2015
2018
2022
Year


FISHERY (whole catch)
Mean length
40
38
36
34
32
30
2012
2014
2016
2018
2020
2022
Year

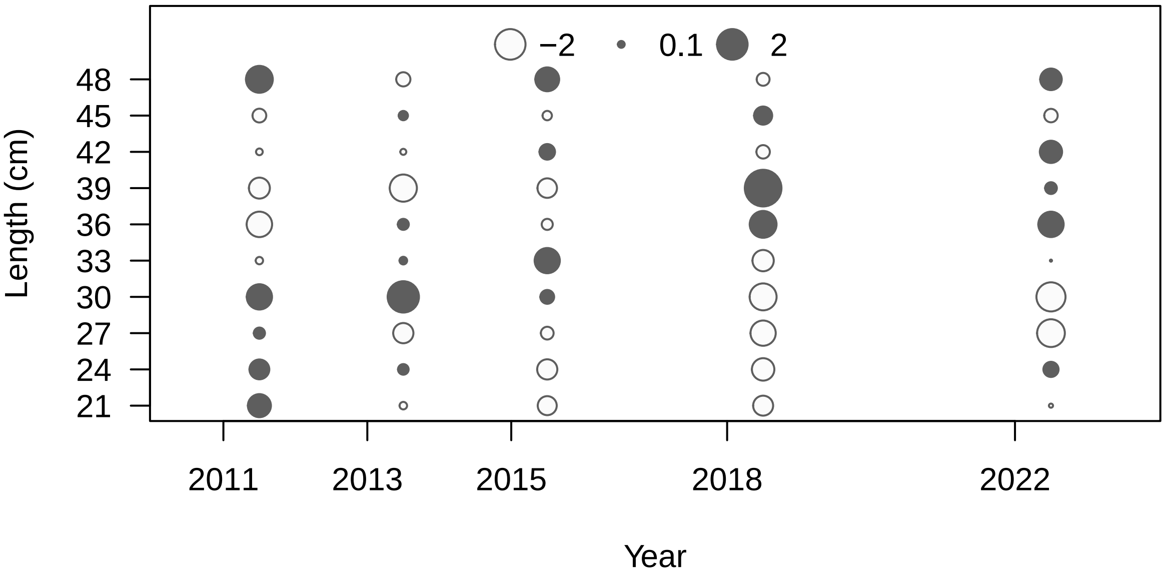

FISHERY
Sum of N input=314
Sum of N adj.=226.1
Proportion
0.00
0.05
0.10
0.15
0.20
0.25
20
25
30
35
40
45
50
Length (cm)



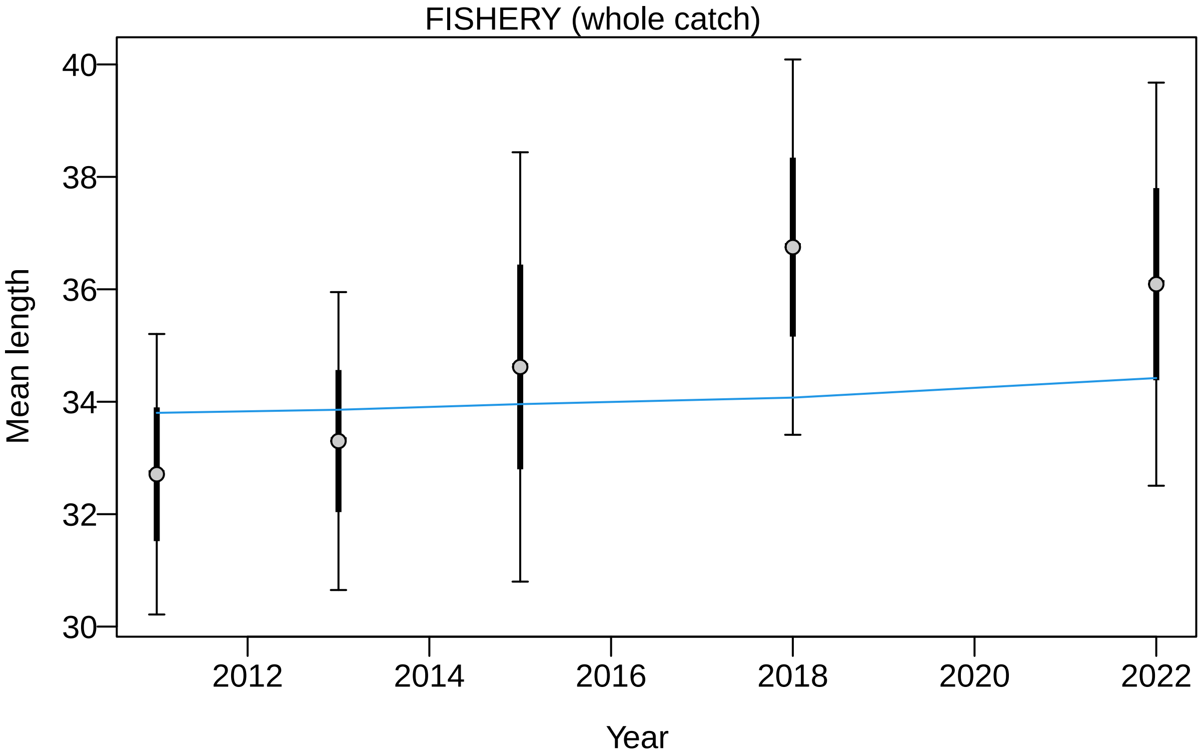
FISHERY
−2
0.1
2
Length (cm)
48
45
42
39
36
33
30
27
24
21
2013
2015
2018
2022
Year

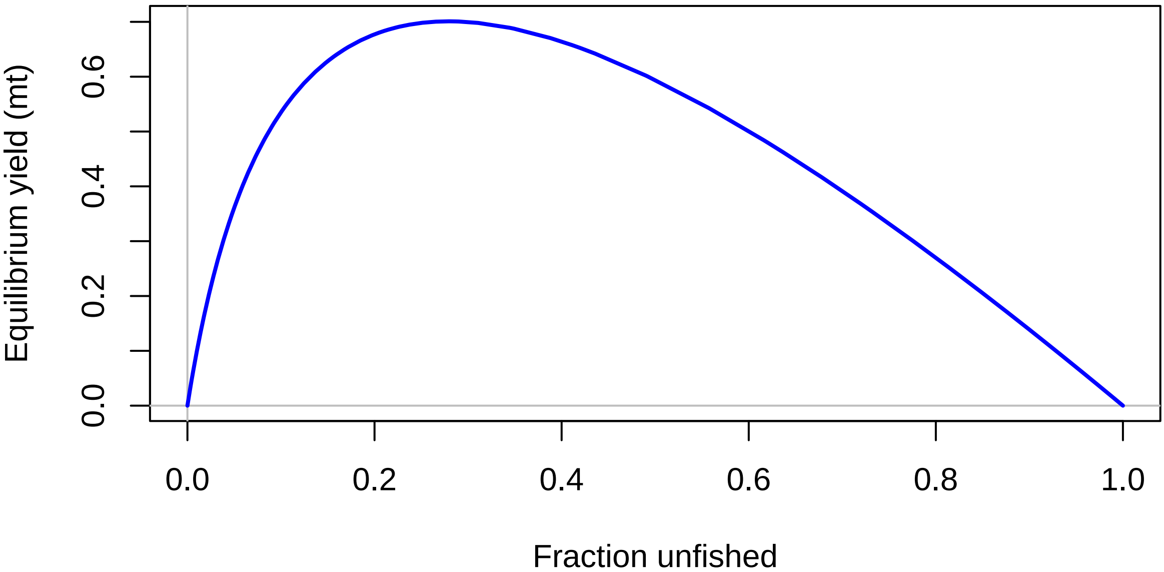


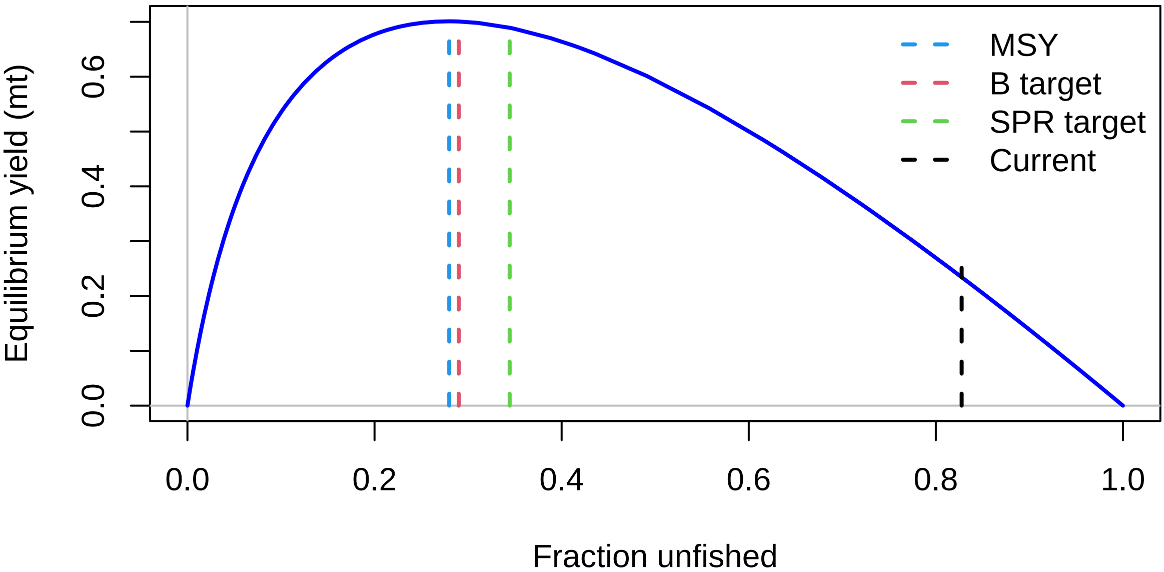
Proportion
2011
N input=94
N adj.=67.6
2013
N input=83
N adj.=59.7
2015
N input=40
N adj.=28.9
2018
N input=52
N adj.=37.5
2022
N input=45
N adj.=32.5
0.0
0.1
0.2
0.3
0.4
20
25
30
35
40
45
50
Length (cm)

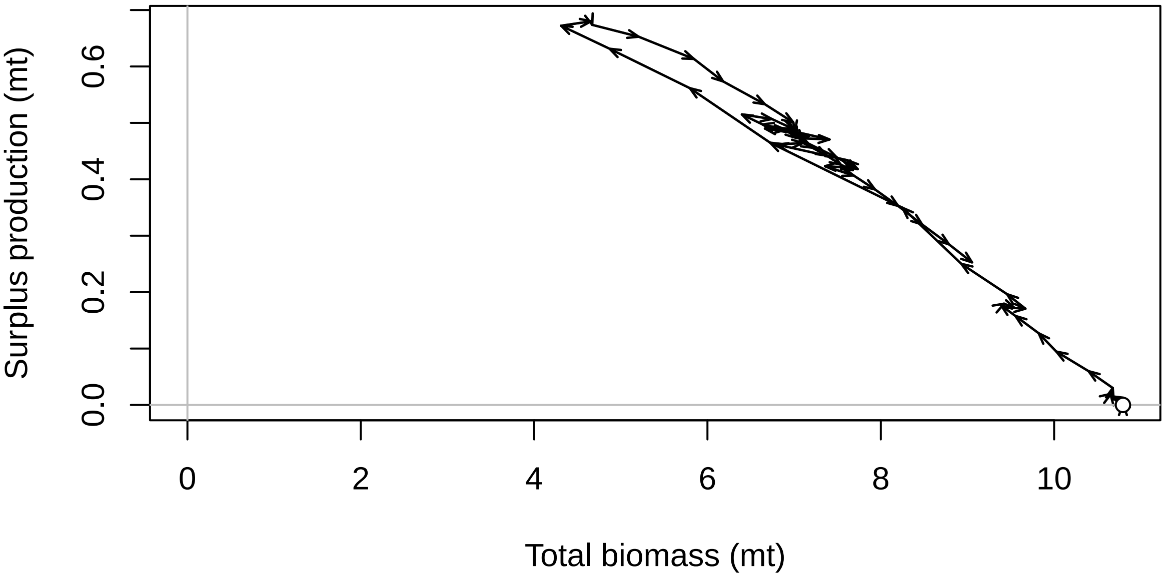


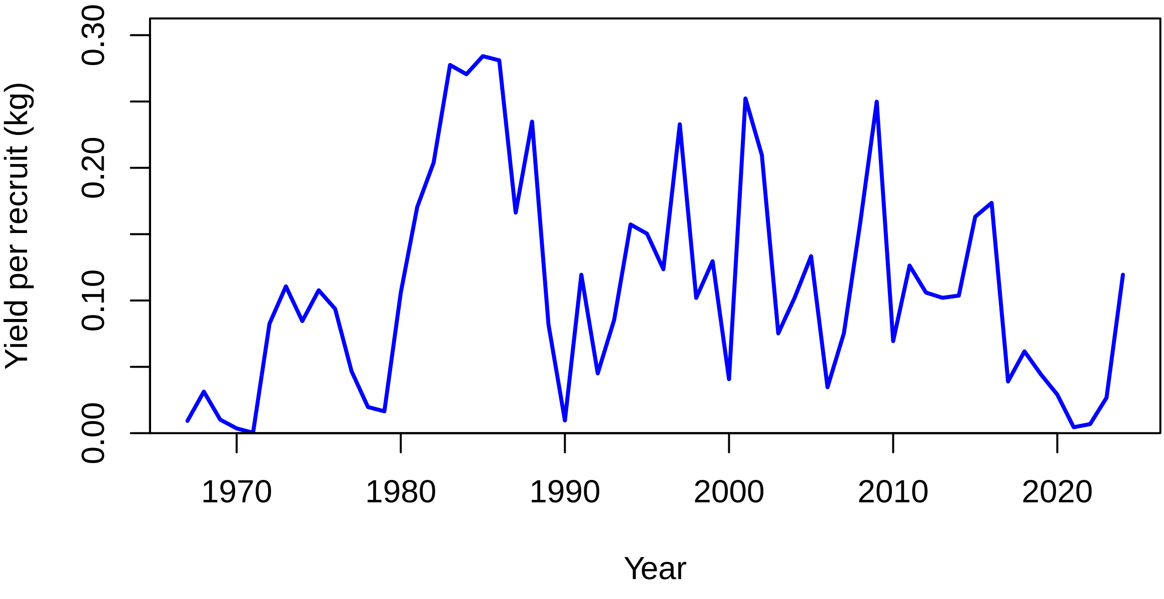
−2
0.1
2
48
45
42
39
36
33
30
27
24
21
Length (cm)
2011
2013
2015
2018
2022
Year


FISHERY (whole catch)
Mean length
40
38
36
34
32
30
2012
2014
2016
2018
2020
2022
Year

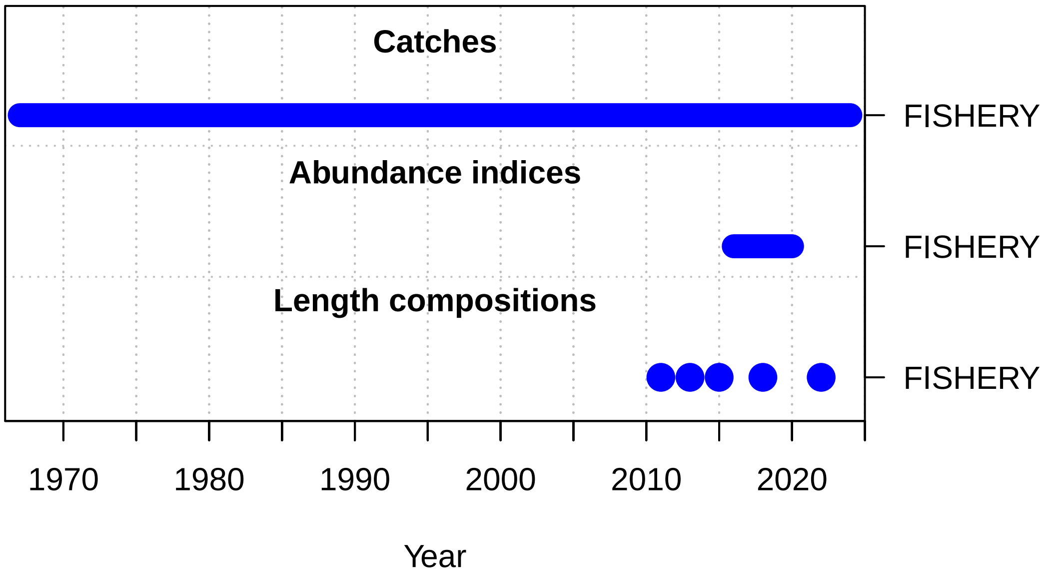

Equilibrium yield (mt)
0.0
0.2
0.4
0.6
0.0
0.2
0.4
0.6
0.8
1.0
Fraction unfished

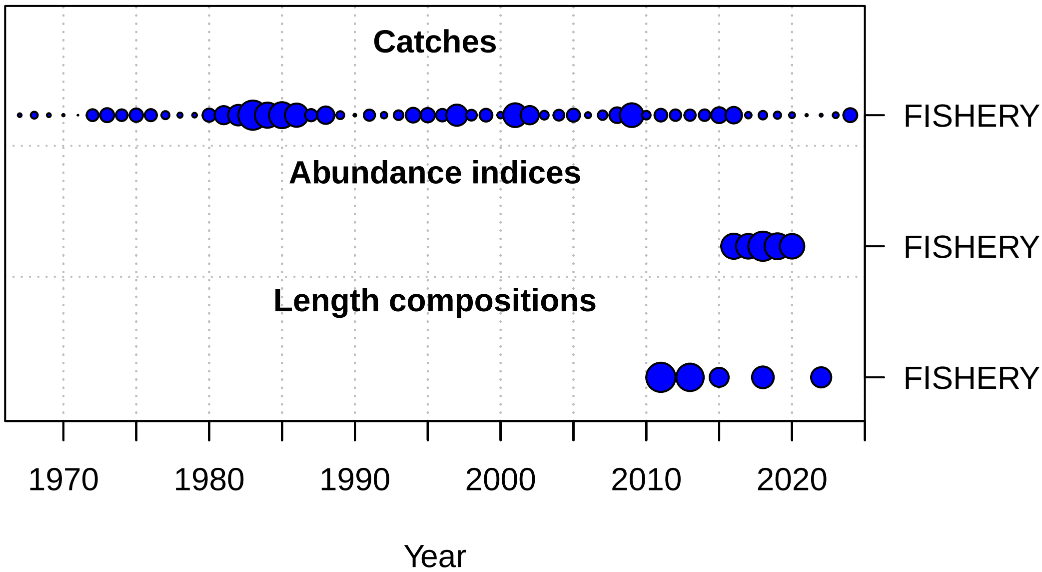

MSY
B target
SPR target
Current
Equilibrium yield (mt)
0.0
0.2
0.4
0.6
0.0
0.2
0.4
0.6
0.8
1.0
Fraction unfished


Surplus production (mt)
0.0
0.2
0.4
0.6
0
2
4
6
8
10
Total biomass (mt)

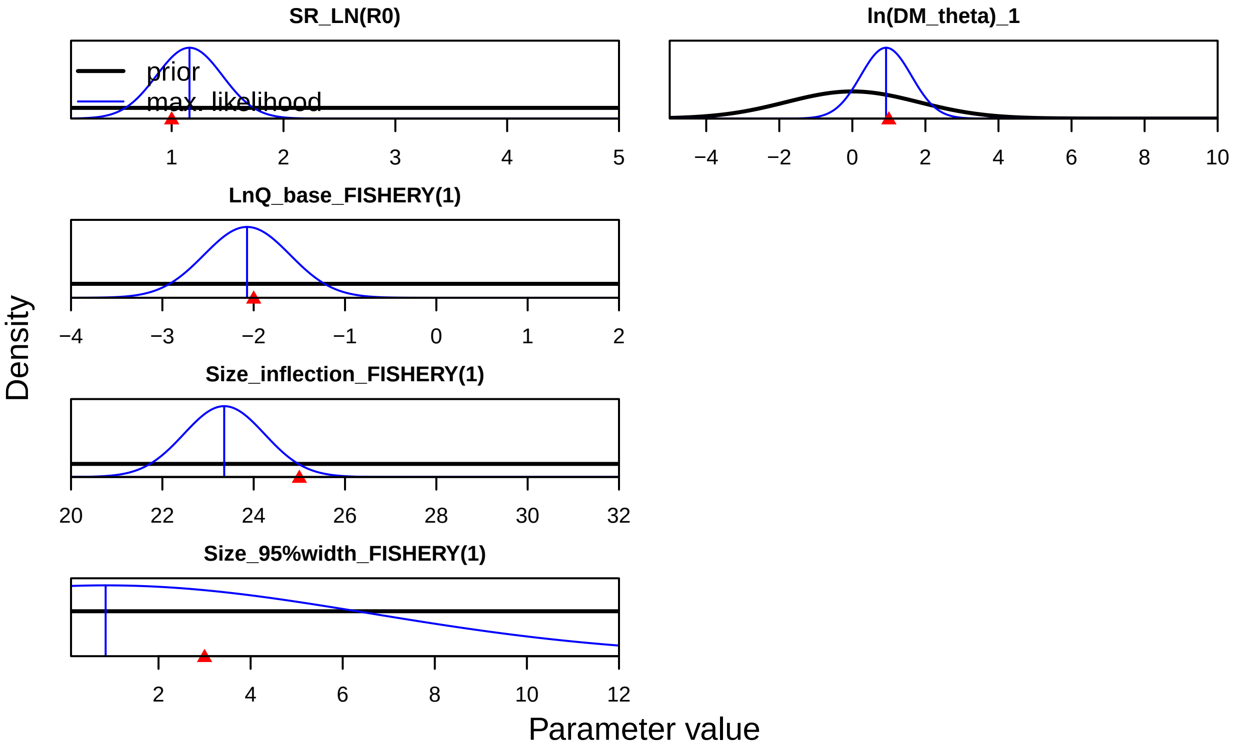


Yield per recruit (kg)
0.00
0.10
0.20
0.30
1970
1980
1990
2000
2010
2020
Year


Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year



Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year


SR_LN(R0)
prior
max. likelihood
1 2 3 4 5
ln(DM_theta)_1
−4 −2 0 2 4 6 8 10
LnQ_base_FISHERY(1)
−4 −3 −2 −1 0 1 2
Size_inflection_FISHERY(1)
20 22 24 26 28 30 32
Size_95%width_FISHERY(1)
2 4 6 8 10 12
Density
Parameter value